

AW-DPCNN Based Multi-Representation Supervised Learning for Mechanical Signal Fault Diagnosis

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Abstract—

Index Terms—Mechanical signal, Gramian angular difference field (GADF), Mel spectrograms, multi-representation, supervised learning, multi-scale convolution, pulse coupled neural network (PCNN).

I. INTRODUCTION

Despite these advances, most existing approaches rely on a single acoustic representation, such as Mel spectrograms or STFT-based spectrograms, which limits their ability to comprehensively capture the complex and complementary fault-related information embedded in transformer signals. Although several studies have attempted to improve performance through multi-representation fusion, these fusion strategies typically employ fixed or heuristic combination schemes and lack the capability to adaptively adjust the contribution of different representations [1], [2]. Furthermore, conventional deep learning models for fault diagnosis often lack explicit mechanisms to exploit complementary heterogeneous feature representations, limiting their robustness and interpretability in complex transformer fault diagnosis tasks.

To overcome the above limitations, we proposed an adaptive weighted dual-channel pulse-coupled neural network (AW-DPCNN) to enhance representation capability and a multi-scale channel attention enhanced VGG16 (MSCA-VGG16) network to improve model robustness. The main contributions of this work can be summarized as follows:

- An AW-DPCNN fusion mechanism is proposed to adaptively integrate Mel and Gramian Angular Difference Field (GADF) representations, enabling more comprehensive modeling of transformer acoustic characteristics.
- To further exploit the complementary information of the fused representation and enhance multi-scale feature sensitivity as well as channel discrimination, an MSCA-VGG16 network is constructed.

The overall framework is shown in Fig. 1. The remainder of this paper is organized as follows. Section II reviews related work on acoustic signal representations, PCNN-based fusion methods, and deep learning for fault diagnosis. Section III presents the proposed methodology, including the AW-DPCNN fusion mechanism and the MSCA-VGG16 classification network. Section IV describes the experimental setup and performance evaluation. Finally, conclusions are drawn in Section V.

II. RELATED WORK

A. Acoustic Signal Representations for Fault Diagnosis

Despite these advances, individual time–frequency or temporal encoding representations each capture only partial aspects

of the acoustic signal. Mel spectrograms primarily reflect local spectral energy distributions but may not fully describe global temporal correlations, while GAF-based representations encode temporal dependencies but lack explicit frequency-domain information. This motivates the development of multi-representation fusion strategies that can jointly exploit complementary signal characteristics.

B. PCNN and Multi-Modal Fusion

Pulse-coupled neural networks (PCNNs) are neural models originally developed to simulate the synchronous firing behavior observed in the mammalian visual cortex [3], [4]. Unlike conventional convolutional neural networks, PCNNs employ a pulse-based activation mechanism in which neurons interact through local coupling and dynamic thresholds [5]. Owing to these characteristics, PCNNs exhibit strong capability in preserving structural information while suppressing background interference, and have been widely applied in image segmentation, edge detection, and feature enhancement tasks [6], [7], [8].

However, conventional PCNNs are typically driven by a single external stimulus, which limits their ability to effectively exploit complementary information from multiple representations. To address this limitation, dual-channel PCNN (DPCNN) architectures have been introduced to enable multi-representation fusion by incorporating multiple external stimuli within the neuronal feeding mechanism [9], [10]. Through this mechanism, heterogeneous feature maps can be jointly processed, thereby facilitating the integration of information from different representations. Nevertheless, the fusion process in such architectures generally relies on fixed or predefined coupling strategies, making it difficult to adaptively regulate the relative contributions of different channels. As a result, the model may not effectively emphasize discriminative representations while suppressing less relevant ones, thereby limiting the effectiveness of multi-representation fusion.

C. Deep Learning for Fault Diagnosis

With the rapid advancement of deep learning techniques, a variety of neural network architectures have been applied to transformer fault identification and rotating machinery fault diagnosis, demonstrating superior automatic feature extraction capability compared with traditional handcrafted methods [11], [12]. Early studies commonly adopted baseline convolutional neural networks as benchmark architectures for acoustic and vibration fault classification, verifying the feasibility of end-to-end feature learning for fault diagnosis tasks [13]. Subsequently, classical deep convolutional backbones, including AlexNet

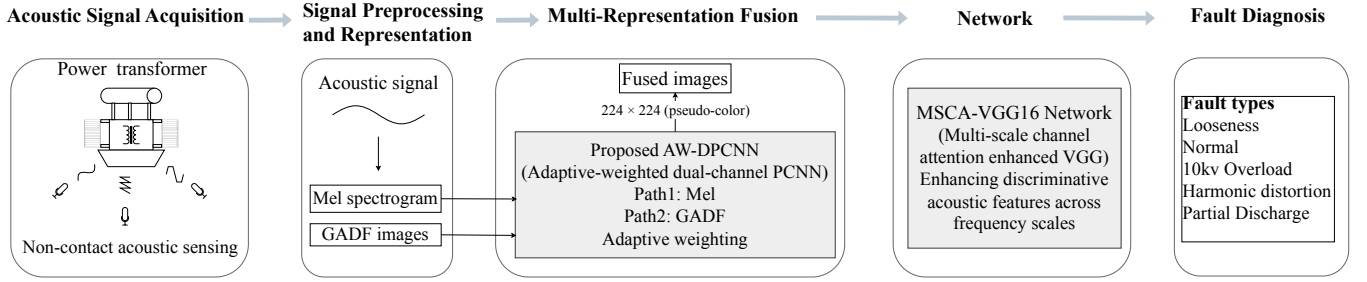


Fig. 1: Overall framework of the proposed transformer acoustic diagnosis method. The acoustic signal is first converted into Mel spectrogram and GADF images. These heterogeneous representations are then adaptively fused through the proposed AW-DPCNN mechanism. Finally, the fused feature is fed into the MSCA-VGG16 network for fault diagnosis.

[14], VGG16 [15], and ResNet18 [16], were introduced into intelligent fault diagnosis frameworks and achieved notable improvements in hierarchical feature representation and classification accuracy across machinery health monitoring applications [17], [18], [19]. More recently, advanced architectures such as ConvNeXt-Tiny, which incorporate modernized convolutional design principles, have further enhanced representation learning capability and demonstrated competitive performance in fault diagnosis and image-based classification tasks [20]. In addition, publicly available benchmark datasets, such as the CWRU bearing fault dataset, have been extensively employed to evaluate the effectiveness and generalization ability of deep learning-based diagnostic models, serving as a standard reference in fault diagnosis research [21], [22].

Despite these advances, most existing deep learning approaches remain purely data-driven and primarily rely on a single representation. Consequently, their diagnostic performance strongly depends on large-scale labeled datasets and is vulnerable to degradation under acoustic noise, operating condition variations, and signal inconsistencies commonly encountered in practical transformer monitoring scenarios. Moreover, conventional deep learning models often lack explicit mechanisms to exploit complementary heterogeneous feature representations, limiting their robustness in complex fault diagnosis tasks.

III. METHODOLOGY

A. Overview

The proposed framework comprises two sequential stages: multi-representation acoustic signal fusion and fused representation classification, as illustrated in Fig. 1. In the fusion stage, the raw acoustic signal is first transformed into two complementary representations—the Mel spectrogram and the Gramian Angular Difference Field (GADF) image—to capture time–frequency energy distributions and global temporal correlation structures, respectively. These heterogeneous representations are then adaptively integrated through the proposed AW-DPCNN mechanism, which employs contrast-guided adaptive weighting and dual-channel pulse-coupled dynamics to produce a unified fused representation. In the classification stage, the fused representation is fed into the MSCA-VGG16 network, which incorporates multi-scale convolution and channel attention to enhance feature discrimination, followed by a compact embedding head for robust fault classification.

Formally, let $\mathbf{x} \in \mathbb{R}^L$ denote a raw acoustic signal segment of length L . The signal is first mapped to two heterogeneous representations as

$$\mathbf{S}_{\text{Mel}} = \mathcal{T}_{\text{Mel}}(\mathbf{x}), \quad \mathbf{S}_{\text{GADF}} = \mathcal{T}_{\text{GADF}}(\mathbf{x}), \quad (1)$$

where $\mathcal{T}_{\text{Mel}}(\cdot)$ and $\mathcal{T}_{\text{GADF}}(\cdot)$ denote the Mel spectrogram and GADF transformation operators, respectively, both producing pseudo-color RGB images of size 224×224 . The two representations are then fused via the AW-DPCNN operator $\mathcal{F}_{\text{AW}}(\cdot, \cdot)$ as

$$\mathbf{S}_{\text{fused}} = \mathcal{F}_{\text{AW}}(\mathbf{S}_{\text{Mel}}, \mathbf{S}_{\text{GADF}}; \Theta_{\text{AW}}), \quad (2)$$

where Θ_{AW} denotes the set of AW-DPCNN hyperparameters. Finally, the fused representation is passed through the MSCA-VGG16 classifier $\mathcal{C}(\cdot)$ to produce the predicted fault class probabilities

$$\hat{\mathbf{y}} = \mathcal{C}(\mathbf{S}_{\text{fused}}; \Theta_{\text{CLS}}), \quad (3)$$

where Θ_{CLS} denotes the trainable parameters of the classification network and $\hat{\mathbf{y}} \in \mathbb{R}^K$ is the predicted probability vector over K fault categories. The following subsections detail each component of this pipeline.

B. Acoustic Signal Representation

1) *Mel Spectrogram*: Direct time-domain analysis is therefore insufficient to capture discriminative fault-related information. To address this issue, the Mel spectrogram is adopted as a perceptually motivated time–frequency representation that aligns with human auditory characteristics [23]. The construction of a Mel spectrogram involves three sequential steps. First, the raw acoustic signal $x(t)$ is segmented into overlapping frames and transformed into the frequency domain via the Short-Time Fourier Transform (STFT), yielding a linear-frequency spectrogram. Second, the resulting power spectra are projected onto a Mel-scale filter bank, where the mapping from linear frequency f to Mel frequency f_{mel} is defined as

$$f_{\text{mel}} = 2595 \log_{10} \left(1 + \frac{f}{700} \right), \quad (4)$$

where f denotes the original frequency in Hertz. Finally, logarithmic amplitude compression is applied to reduce the

dynamic range and enhance the visibility of low-energy components. Consequently, the Mel spectrogram forms a compact two-dimensional representation $S_{\text{Mel}}(t, f_{\text{mel}})$, where t and f_{mel} denote the time frame index and the Mel-frequency bin, respectively. This representation effectively captures the spectral energy distribution and its temporal evolution, making it suitable for analyzing transformer acoustic emissions under stationary or slowly varying operating conditions. Fig. 2(a) illustrates an example of a raw transformer acoustic signal and its corresponding Mel spectrogram, where fault-induced spectral energy redistribution can be clearly observed.

However, the Mel spectrogram exhibits several inherent limitations. First, the logarithmic compression, while beneficial for dynamic range reduction, also amplifies low-level background noise, making the representation sensitive to acoustic interference commonly encountered in substation environments. Second, the Mel-scale filter bank inherently applies a nonlinear frequency compression that reduces spectral resolution at higher frequencies, potentially obscuring fine-grained high-frequency fault signatures. More fundamentally, the Mel spectrogram primarily characterizes local time–frequency energy distributions and does not explicitly encode the global temporal correlation structure of the acoustic signal. As a result, fault patterns that manifest as long-range temporal dependencies rather than localized spectral anomalies may not be adequately captured by Mel spectrograms alone.

In this study, the Mel spectrogram is computed with the following parameter settings: the STFT employs an FFT window size of $n_{\text{fft}} = 2048$ and a hop length of 256 samples; the Mel filter bank consists of $n_{\text{mels}} = 128$ triangular filters spanning up to a maximum frequency of $f_{\text{max}} = 8000$ Hz. These parameters are chosen to balance time–frequency resolution for the analysis of transformer acoustic signals sampled at 44.1 kHz. After conversion to a decibel scale via logarithmic compression, the resulting grayscale representation is mapped to a pseudocolor image of size 224×224 using the Viridis colormap for compatibility with standard CNN input dimensions.

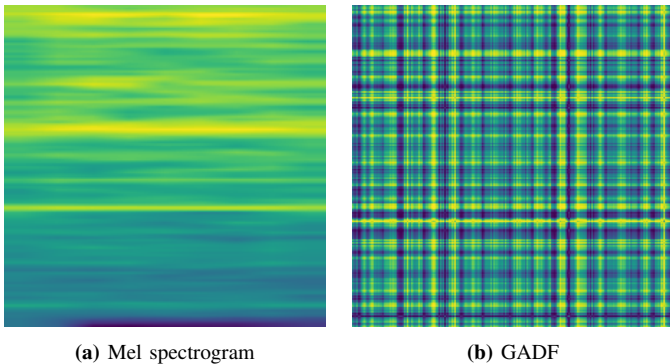


Fig. 2: Transformer acoustic signal representations.

2) *Gramian Angular Field:* To address the limitations of the Mel spectrogram in encoding global temporal structures, the Gramian Angular Field (GAF) is introduced as a complementary representation. The GAF is a method that transforms one-dimensional time-series signals into two-dimensional images while preserving and encoding rich temporal correlation

information [24]. It was originally proposed to facilitate time-series classification through computer vision techniques [25], [26]. The construction of a GAF representation involves three steps:

- 1) The time series $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$ is first normalized to the interval $[-1, 1]$ using min–max normalization, producing a normalized sequence $\tilde{\mathbf{X}} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n\}$, where each $\tilde{x}_i$ is computed as

$$\tilde{x}_i = \frac{(x_i - \min \mathbf{X}) + (x_i - \max \mathbf{X})}{\max \mathbf{X} - \min \mathbf{X}}. \quad (5)$$

- 2) The normalized values $\tilde{x}_i$ are then encoded as angular coordinates in a polar coordinate system. Specifically, each $\tilde{x}_i$ is mapped to a polar angle ϕ_i , while the corresponding radial coordinate r_i represents the temporal order of the sequence. This transformation is defined as

$$\begin{cases} \phi_i = \arccos(\tilde{x}_i), & -1 \leq \tilde{x}_i \leq 1, \tilde{x}_i \in \tilde{\mathbf{X}} \\ r_i = \frac{i}{n}, & i = 1, \dots, n \end{cases}, \quad (6)$$

where ϕ_i denotes the polar angle of the i -th point and r_i represents its normalized temporal position. This polar encoding ensures that the temporal ordering of the signal is preserved through the radial coordinate, while signal amplitude variations are encoded as angular deviations.

- 3) Finally, the GAF is constructed by aggregating the phase angles through either summation or difference operations, yielding two complementary variants: the Gramian Angular Summation Field (GASF), denoted as $\mathbf{G}_s$, and the Gramian Angular Difference Field (GADF), denoted as $\mathbf{G}_d$. Their formulations are defined as

$$\mathbf{G}_s(i, j) = \cos(\phi_i + \phi_j), \quad i, j = 1, \dots, n \quad (7)$$

$$\mathbf{G}_d(i, j) = \sin(\phi_i - \phi_j), \quad i, j = 1, \dots, n. \quad (8)$$

The GAF representation offers several advantages that complement the Mel spectrogram. First, by encoding pairwise angular relationships between all time points, the GAF captures global temporal correlation structures that extend beyond the local receptive field of STFT-based representations. This property makes it particularly effective for detecting fault-induced irregularities in temporal dynamics, such as those caused by harmonic distortions or partial discharge events. Second, the min–max normalization in (5) inherently suppresses the influence of absolute signal amplitude, making the GAF representation more robust to variations in signal gain and background noise levels compared with raw spectrogram representations. Furthermore, the polar encoding in (6) provides a bijective mapping between the original time series and its angular representation, ensuring that temporal information is preserved without information loss.

In this study, the GADF variant is specifically adopted to complement the Mel spectrogram. Compared with GASF, which encodes angular summation, GADF focuses on angular differences and thereby exhibits higher sensitivity to abrupt temporal variations and asymmetric dynamic patterns [27].

This property is particularly beneficial for transformer acoustic fault diagnosis, where abnormal operating conditions often produce irregular temporal fluctuations in the acoustic signals. As shown in Fig. 2(b), the GADF encodes structured temporal dependencies from acoustic signals under normal operating conditions. The resulting image exhibits regular and symmetric patterns, reflecting stable temporal correlations and consistent signal dynamics. Such characteristics provide discriminative cues for distinguishing normal conditions from fault-induced irregular patterns.

For the experiments in this study, the GADF is computed with a sequence length of $n = 224$, matching the target image dimension. The angular difference variant (GADF) is selected rather than the summation variant (GASF) due to its superior sensitivity to abrupt temporal variations. The resulting GADF matrix is normalized to $[0, 1]$, converted to an 8-bit grayscale image, and subsequently mapped to a pseudo-color representation of size 224×224 using the Viridis colormap. Together, the Mel spectrogram and GADF constitute a complementary representation pair that jointly captures local spectral energy patterns and global temporal correlation structures, providing a comprehensive basis for the subsequent adaptive fusion stage.

C. AW-DPCNN for Multi-Representation Fusion

As discussed in Section II, the Mel spectrogram and GADF provide fundamentally different yet complementary representations of transformer acoustic signals. Specifically, the Mel spectrogram primarily characterizes local time–frequency energy distributions and perceptually relevant spectral patterns, whereas the GADF emphasizes global temporal dependencies and dynamic correlation structures. Owing to their distinct statistical properties and structural characteristics, directly combining these heterogeneous representations using conventional pixel-wise averaging or simple feature concatenation may fail to effectively capture their cross-modal relationships and may even introduce destructive interference, as demonstrated in the ablation study (Section IV). To address this issue, the proposed AW-DPCNN introduces a PCNN-inspired fusion mechanism with three key innovations: (i) dual-channel external stimuli that independently process Mel and GADF representations through representation-specific convolutional kernels; (ii) a contrast-guided adaptive weighting scheme that dynamically adjusts the relative contribution of each representation based on local saliency; and (iii) iterative pulse-coupled dynamics that propagate and reinforce structurally consistent features across multiple iterations while suppressing spurious activations.

The pulse coupling characteristic of PCNN enables dynamic interaction among neighboring feature responses, allowing salient structures shared across Mel and GADF representations to be synchronously enhanced while suppressing irrelevant or inconsistent activations [28]. Such a mechanism is particularly suitable for acoustic representation fusion, where local spectral textures and global temporal correlation patterns coexist across different feature domains [29]. Moreover, the adaptive weighting strategy further regulates the contribution of each representation according to their fault-sensitive responses,

thereby improving feature complementarity and discriminative capability.

Building upon these characteristics, the proposed AW-DPCNN integrates dual-channel processing, adaptive weighting, and multi-scale convolution to effectively model cross-representation dependencies and hierarchical feature interactions between Mel and GADF representations, ultimately enhancing transformer fault diagnosis performance. The key hyperparameters of the AW-DPCNN are summarized in Table III, and the detailed formulation of each component is presented below.

1) *Input Normalization*: To eliminate scale differences between two representations, each input is first normalized to the interval $[0, 1]$ as

$$\tilde{S}_k(i, j) = \frac{S_k(i, j) - \min_{i,j}(S_k(i, j))}{\max_{i,j}(S_k(i, j)) - \min_{i,j}(S_k(i, j)) + \epsilon}, k \in \{1, 2\} \quad (9)$$

where $S_k(i, j)$ denotes the value of the k -th input representation at spatial location (i, j) , corresponding to the Mel spectrogram ($k = 1$) or the GADF image ($k = 2$). The variables i and j denote the pixel indices, and ϵ is a small positive constant preventing division by zero.

2) *Contrast-Guided Adaptive Weighting*: For each representation, the local mean of the normalized representation is computed as

$$\mu_k(i, j) = \frac{1}{|\Omega|} \sum_{(p,q) \in \Omega} \tilde{S}_k(p, q), \quad (10)$$

where Ω denotes the local neighborhood centered at each pixel location, and $|\Omega|$ represents the number of elements in the neighborhood.

The corresponding local contrast is defined as

$$C_k(i, j) = \sqrt{\mathbb{E}_{\Omega} [\tilde{S}_k^2] - \mu_k^2(i, j)}, \quad (11)$$

where $\mathbb{E}[\cdot]$ denotes local averaging within Ω .

Local contrast operators $C_1(i, j)$ and $C_2(i, j)$ are computed to quantify the saliency of Mel and GADF representations at each spatial location. To emphasize the dominant representation, a contrast amplification factor γ is introduced to construct adaptive fusion weights as follows

$$\beta_1(i, j) = \frac{\gamma C_1(i, j)}{\gamma C_1(i, j) + C_2(i, j) + \epsilon}, \quad (12)$$

$$\beta_2(i, j) = \frac{C_2(i, j)}{\gamma C_1(i, j) + C_2(i, j) + \epsilon}, \quad (13)$$

the above formulation ensures $\beta_1(i, j) + \beta_2(i, j) \approx 1$, enabling contrast-driven dynamic weighting.

3) *Dual-Channel PCNN*: Let $L^n(i, j)$, $T^n(i, j)$, and $Y^n(i, j)$ denote the linking term, dynamic threshold, and binary pulse output, respectively, where n denotes the current iteration and $n \in [1, N]$, with N being the total number of iterations. The neural states are initialized as

$$L^0(i, j) = 0, \quad T^0(i, j) = 0, \quad Y^0(i, j) = 0. \quad (14)$$

At the n -th iteration, the external stimulus from two channels is defined as

$$\mathbf{F}_1^n(i, j) = (\mathbf{W}_1 * \mathbf{Y}^{n-1})(i, j) + \tilde{\mathbf{S}}_1(i, j) \quad (15)$$

$$\mathbf{F}_2^n(i, j) = (\mathbf{W}_2 * \mathbf{Y}^{n-1})(i, j) + \tilde{\mathbf{S}}_2(i, j), \quad (16)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ denote the convolutional kernels corresponding to the Mel and GADF channels, respectively, and $*$ represents the convolution operation. Considering the distinct structural characteristics of the two representations, representation-specific convolutional kernels are introduced in the dual-channel architecture. Specifically, stripe-shaped kernels are applied to the Mel channel to emphasize frequency continuity and harmonic structures, while symmetric kernels are used for the GADF channel to capture temporal periodicity and global correlation patterns. In this study, $\mathbf{W}_1$, $\mathbf{W}_2$, and $\mathbf{M}$ are structural design choices grounded in the physical interpretation of each representation, not free hyperparameters. The internal linking state is updated by

$$\mathbf{L}^n(i, j) = e^{-\alpha_L} \mathbf{L}^{n-1}(i, j) + (\mathbf{M} * \mathbf{Y}^{n-1})(i, j), \quad (17)$$

where α_L is the decay coefficient and $\mathbf{M}$ represents the coupling matrix.

The total neuron input is then given by

$$\mathbf{U}^n(i, j) = \mathbf{L}^n(i, j) \left(1 + \beta_1 \mathbf{F}_1^n(i, j)\right) \left(1 + \beta_2 \mathbf{F}_2^n(i, j)\right) + \sigma, \quad (18)$$

where σ is a stabilization constant.

The pulse output is determined by a threshold comparison as follows

$$\mathbf{Y}^n(i, j) = \begin{cases} 1, & \mathbf{U}^n(i, j) > \mathbf{T}^{n-1}(i, j) \\ 0, & \text{otherwise} \end{cases}. \quad (19)$$

The dynamic threshold evolves according to

$$\mathbf{T}^n(i, j) = e^{-\alpha_T} \mathbf{T}^{n-1}(i, j) + V_T \mathbf{Y}^n(i, j), \quad (20)$$

where α_T controls threshold decay and V_T is the amplification coefficient.

4) *Fused Representation*: The accumulated neural response after N iterations is

$$\mathbf{U}_{\text{sum}}(i, j) = \sum_{n=1}^N \mathbf{U}^n(i, j), \quad (21)$$

and the final fused representation is obtained via normalization as

$$\mathbf{F}(i, j) = \frac{\mathbf{U}_{\text{sum}}(i, j) - \min_{i,j} \mathbf{U}_{\text{sum}}(i, j)}{\max_{i,j} \mathbf{U}_{\text{sum}}(i, j) - \min_{i,j} \mathbf{U}_{\text{sum}}(i, j) + \epsilon}. \quad (22)$$

Through the proposed adaptive fusion mechanism, Mel and GADF representations are no longer combined with fixed weights. Instead, the neuron state is dynamically modulated in a demand-driven manner, i.e., regions dominated by spectral variations are mainly governed by Mel, while regions with stronger temporal correlations rely more on GADF. This adaptive strategy effectively suppresses fusion artifacts while enhancing the discriminative capability of acoustic representations.

The overall architecture of the proposed AW-DPCNN is illustrated in Fig. 3. By integrating Mel spectral perception and GADF temporal encoding within a unified adaptive fusion framework, the proposed AW-DPCNN produces robust and discriminative acoustic representations for transformer fault diagnosis. The qualitative effectiveness of the proposed adaptive fusion mechanism is further analyzed through feature visualization in Section IV.

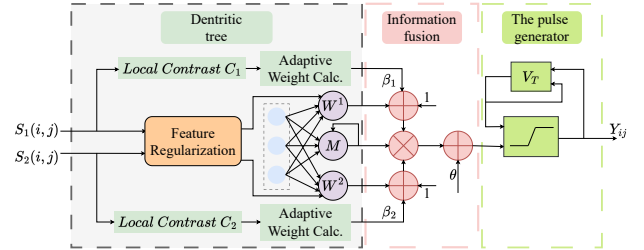


Fig. 3: Architecture of the proposed adaptive weighted dual-channel PCNN (AW-DPCNN) for multi-representation acoustic signal fusion.

D. MSCA-VGG16 for Fused Representation Classification

1) *Design Philosophy and Overview*: The fused representations $\mathbf{S}_{\text{fused}}$ generated by AW-DPCNN simultaneously encode local time–frequency energy variations from Mel spectrograms and global temporal correlation structures from GADF encoding. These heterogeneous features exhibit distinct statistical distributions: Mel-derived components emphasize localized spectral textures, while GADF-derived components reveal long-range structural dependencies. Conventional convolutional networks with fixed receptive fields and uniform channel treatment are ill-suited to jointly exploit such complementary information. To address this, we develop a Multi-Scale Channel Attention enhanced network (MSCA-VGG16) built upon three design principles: (i) *multi-scale receptive field alignment* to capture fault patterns spanning different temporal and spectral granularities; (ii) *dynamic channel recalibration* via squeeze-and-excitation to suppress noise-dominated frequency bands while amplifying fault-sensitive responses; and (iii) *compact discriminative embedding* to project high-dimensional convolutional features into a lower-dimensional space that enhances intra-class compactness and inter-class separability.

The overall architecture is a modular composition of a convolutional backbone $\mathcal{B}(\cdot)$, a multi-scale channel attention module $\mathcal{M}(\cdot)$, global pooling $\mathcal{P}(\cdot)$, an embedding projection $\mathcal{E}(\cdot)$, and a linear classifier $\mathcal{F}_{cls}(\cdot)$. The forward mapping from input $\mathbf{X}$ to predicted class probabilities $\hat{\mathbf{y}}$ is

$$\hat{\mathbf{y}} = \mathcal{F}_{cls}(\mathcal{E}(\mathcal{P}(\mathcal{M}(\mathcal{B}(\mathbf{X}))))), \quad (23)$$

where $\mathbf{X} \in \mathbb{R}^{H_0 \times W_0 \times 3}$ denotes the AW-DPCNN fused pseudo-color image of spatial size $H_0 \times W_0$. The following subsections detail each component.

2) *Convolutional Backbone*: The backbone $\mathcal{B}(\cdot)$ serves as a hierarchical feature extractor that maps the fused input image to a rich spatial feature representation. It consists of a stack of convolutional stages with progressively increasing channel dimensionality and decreasing spatial resolution, implemented

through strided convolutions or pooling operations. Formally, the backbone can be any deep convolutional architecture that accepts an RGB image and produces a feature map

$$\mathbf{F} = \mathcal{B}(\mathbf{X}), \quad \mathbf{F} \in \mathbb{R}^{H \times W \times C}, \quad (24)$$

where H , W , and C denote the spatial height, width, and number of channels of the output feature map, respectively. The spatial dimensions H and W are determined by the cumulative downsampling factor of the backbone, while C corresponds to the channel capacity of the final convolutional stage.

In this study, VGG16 with batch normalization (VGG16-BN) [15] is selected as the primary convolutional backbone for the MSCA framework. The selection is guided by three interrelated considerations: architectural compatibility with the auxiliary modules, experimental interpretability, and the avoidance of confounding mechanisms.

Architectural compatibility. The MSCA framework attaches three auxiliary modules—the multi-scale convolution (MS) module, the squeeze-and-excitation channel attention (CA) module, and the compact embedding head (EH)—onto the backbone’s output feature map. This imposes two structural requirements on the backbone: (i) a linear, single-path feedforward topology that permits clean insertion of auxiliary modules without routing around residual branches, and (ii) a sufficiently rich output channel capacity to support the MS module’s parallel branches without channel competition. VGG16-BN satisfies both requirements: its architecture is a pure sequential stack of 3×3 convolutional blocks with no skip connections, identity shortcuts, or built-in attention mechanisms, providing a uniform insertion surface for the MS, CA, and EH modules. Its final convolutional stage outputs 512-channel feature maps at 7×7 spatial resolution, offering ample representational headroom for the three parallel multi-scale branches to extract complementary patterns. In contrast, architectures with residual connections introduce branching topologies that complicate the definition of insertion points: the MS module’s parallel branches would require separate identity paths, conflating the contribution of multi-scale processing with that of the backbone’s own skip structure. Architectures with built-in SE modules (e.g., EfficientNet’s MBConv blocks) create redundancy with the CA module, making it impossible to attribute performance gains unambiguously. Vision Transformers lack the convolutional inductive bias and spatial hierarchy that the MS module relies upon to capture patterns at different spatial granularities.

Experimental interpretability. A central objective of this study is to evaluate the effectiveness of the MS, CA, and EH modules in enhancing feature discrimination for fused acoustic representations. Isolating the contribution of these modules from the backbone’s inherent properties requires a backbone that provides a neutral starting point—that is, an architecture with no attention, no multi-scale processing, and no skip connections. VGG16-BN, being the most canonical deep convolutional architecture, provides precisely such a baseline. Every performance gain can therefore be unambiguously attributed to the proposed modules rather than to synergies with the backbone’s own design. Furthermore, VGG16 has a single well-defined variant with batch normalization, eliminating the

degrees of freedom associated with selecting among multiple variants (e.g., ResNet-18/34/50/101) and thereby enhancing reproducibility.

Avoidance of confounding mechanisms. ResNet’s skip connections inherently provide a form of implicit multi-scale gradient flow that could partially mask the benefit of the MS module. EfficientNet’s MBConv blocks already contain SE operations, rendering the CA module redundant and preventing a clean ablation of channel attention. ViT’s self-attention mechanism operates globally by design, which conflicts with the locally-grounded multi-scale receptive fields of the MS branches. VGG16-BN possesses none of these mechanisms, serving as a “blank canvas” on which the effect of each proposed module can be independently assessed.

To validate this selection empirically, a comparative experiment is conducted in Section IV-C, where the MSCA framework is instantiated with several alternative backbones, including EfficientNet-B0, MobileNetV3-Small, ConvNeXt-Tiny, and ViT. The results confirm that VGG16-BN, when augmented with the MS, CA, and EH modules, consistently achieves the most favorable accuracy–efficiency trade-off on the target diagnostic task, corroborating the architectural analysis presented above.

3) Multi-Scale Convolution Module: Fault signatures in fused acoustic representations manifest at multiple scales: fine-grained local anomalies (e.g., impulsive spectral lines), mid-range harmonic patterns, and long-range structural correlations across time and frequency. A single receptive field cannot simultaneously capture all three. To align the network’s perceptual field with this multi-scale nature, we introduce a multi-scale convolution (MS) module $\psi_{\text{MS}}(\cdot)$ comprising S parallel branches with different effective receptive fields:

$$\mathbf{F}_{\text{ms}} = \psi_{\text{MS}}(\mathbf{F}) = \sum_{i=1}^S \psi_i(\mathbf{F}; \Theta_i), \quad (25)$$

where $\psi_i(\cdot; \Theta_i)$ denotes the i -th convolutional branch parameterized by Θ_i . Each branch employs a distinct kernel configuration to capture patterns at a specific scale: a standard small-kernel convolution for local fine-grained textures, a larger-kernel convolution for mid-range structural features, and a dilated convolution that expands the receptive field without additional parameters for long-range dependencies. Each branch preserves the channel dimensionality C of the backbone output.

The outputs are fused via element-wise summation, which encourages each branch to specialize in complementary scales. The aggregated feature map is normalized and activated:

$$\mathbf{F}_{\text{act}} = \delta(\text{BN}(\mathbf{F}_{\text{ms}})), \quad (26)$$

where $\delta(\cdot) = \max(0, \cdot)$ is the ReLU activation and $\text{BN}(\cdot)$ denotes batch normalization. The specific kernel sizes, dilation rates, and the number of branches S are provided as hyperparameter configurations in the experiment section (Section IV).

4) Squeeze-and-Excitation Channel Attention: Different frequency bands in the fused representation carry unequal diagnostic information. Some channels encode fault-critical spectral components, while others are dominated by background noise or redundant textures. To dynamically recalibrate

channel-wise responses, we adopt a squeeze-and-excitation (SE) mechanism [30] after the multi-scale fusion stage.

Let $\mathbf{F}_{\text{act}} \in \mathbb{R}^{H \times W \times C}$ be the input to the SE module. The *squeeze* operation aggregates spatial information into a channel descriptor $\mathbf{z} \in \mathbb{R}^C$ via global average pooling:

$$z_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \mathbf{F}_{\text{act}}(i, j, c), \quad c = 1, \dots, C. \quad (27)$$

The *excitation* operation models channel-wise dependencies through a bottleneck structure:

$$\mathbf{s} = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z})), \quad (28)$$

where $\mathbf{W}_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ are learnable weight matrices, r is the channel reduction ratio controlling the bottleneck capacity, and $\sigma(\cdot)$ is the sigmoid function. The output $\mathbf{s} = [s_1, \dots, s_C]^\top$ represents per-channel attention weights in $[0, 1]$. The refined feature map is obtained by channel-wise rescaling:

$$\mathbf{F}_{\text{att}}(i, j, c) = s_c \cdot \mathbf{F}_{\text{act}}(i, j, c). \quad (29)$$

This mechanism allows the network to emphasize channels containing fault-discriminative information (e.g., frequency bands with strong harmonic signatures) while attenuating noise-dominated channels. The reduction ratio r is specified in the experiment configuration.

5) *Compact Discriminative Embedding Head*: Conventional CNN classifiers often employ large fully-connected layers that are prone to overfitting on small-scale fault diagnosis datasets and introduce substantial parameter redundancy [31]. We replace these with a compact embedding head that projects high-dimensional convolutional features into a low-dimensional discriminative space, promoting intra-class compactness and inter-class separation while reducing the parameter count.

First, the spatially refined feature map $\mathbf{F}_{\text{att}}$ is collapsed via global average pooling and flattened:

$$\mathbf{f}_0 = \text{Flatten}(\text{AvgPool}(\mathbf{F}_{\text{att}})) \in \mathbb{R}^C. \quad (30)$$

A fully-connected projection layer extracts high-level features:

$$\mathbf{f}_1 = \delta(\mathbf{W}_{fc1} \mathbf{f}_0), \quad \mathbf{W}_{fc1} \in \mathbb{R}^{D_1 \times C}, \quad (31)$$

followed by dropout for regularization. A compact embedding layer then projects $\mathbf{f}_1$ into a d -dimensional discriminative space:

$$\mathbf{e} = \delta(\text{BN}(\mathbf{W}_e \mathbf{f}_1)), \quad \mathbf{e} \in \mathbb{R}^d, \quad (32)$$

where $\mathbf{W}_e \in \mathbb{R}^{d \times D_1}$. Batch normalization is applied prior to ReLU to stabilize the feature distribution, and dropout follows the embedding layer. Finally, the class logits are obtained via a linear classifier:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_c \mathbf{e}), \quad \mathbf{W}_c \in \mathbb{R}^{K \times d}, \quad (33)$$

where K is the number of fault categories. The dimensions D_1 (projection layer width) and d (embedding dimensionality), as well as the dropout probability, are treated as hyperparameters whose concrete values are given in Section IV. The entire network is optimized by minimizing the cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = - \sum_{k=1}^K y_k \log \hat{y}_k, \quad (34)$$

where $\mathbf{y} = [y_1, \dots, y_K]^\top$ is the one-hot ground-truth label. In practice, a class-weighted variant $\mathcal{L}_{\text{wCE}}$ is employed to counteract dataset imbalance (Section IV).

6) *Forward Pass Algorithm*: Algorithm 1 summarizes the complete forward pass of the proposed MSCA-VGG16 network from the fused input image to the predicted class probabilities.

Algorithm 1 Forward Pass of MSCA-VGG16

Require: Fused AW-DPCNN image $\mathbf{X} \in \mathbb{R}^{H_0 \times W_0 \times 3}$

- 1: Extract backbone feature map $\mathbf{F}$ using Eq. 24.
- 2: Compute multi-scale feature $\mathbf{F}_{\text{ms}}$ using Eq. 25.
- 3: Apply BatchNorm and ReLU to obtain $\mathbf{F}_{\text{act}}$ using Eq. 26.
- 4: Squeeze spatial information into channel descriptor $\mathbf{z}$ using Eq. 27.
- 5: Compute channel attention weights $\mathbf{s}$ using Eq. 28.
- 6: Obtain channel-refined feature $\mathbf{F}_{\text{att}}$ using Eq. 29.
- 7: Apply global average pooling and flatten to obtain $\mathbf{f}_0$ using Eq. 30.
- 8: Project to high-level features $\mathbf{f}_1$ using Eq. 31, then apply dropout.
- 9: Compute compact embedding $\mathbf{e}$ using Eq. 32, then apply dropout.
- 10: Produce class probabilities $\hat{\mathbf{y}}$ using Eq. 33.

Ensure: Predicted probabilities $\hat{\mathbf{y}} \in \mathbb{R}^K$, optimized via Eq. 34.

The overall architecture of the proposed MSCA-VGG16 network is illustrated in Fig. 5.

E. Training Strategy

The MSCA-VGG16 classification network is trained using a supervised learning protocol with the following configuration. All experiments adopt the AdamW optimizer [?] with an initial learning rate of $\eta = 1 \times 10^{-4}$ and a weight decay of $\lambda = 1 \times 10^{-3}$, which provides adaptive learning rate scaling with decoupled weight regularization for improved generalization. A ReduceLROnPlateau learning rate scheduler is employed, which reduces the learning rate by a factor of 0.5 when the validation F1-score does not improve for 5 consecutive epochs, with a minimum learning rate of 1×10^{-6} . The batch size is set to 32 and the maximum number of training epochs is set to 30.

To mitigate the adverse effects of class imbalance inherent in practical transformer acoustic datasets, a class-weighted cross-entropy loss is adopted. Specifically, the weight w_k for class k is computed as

$$w_k = \frac{N_{\text{total}}}{K \cdot N_k}, \quad (35)$$

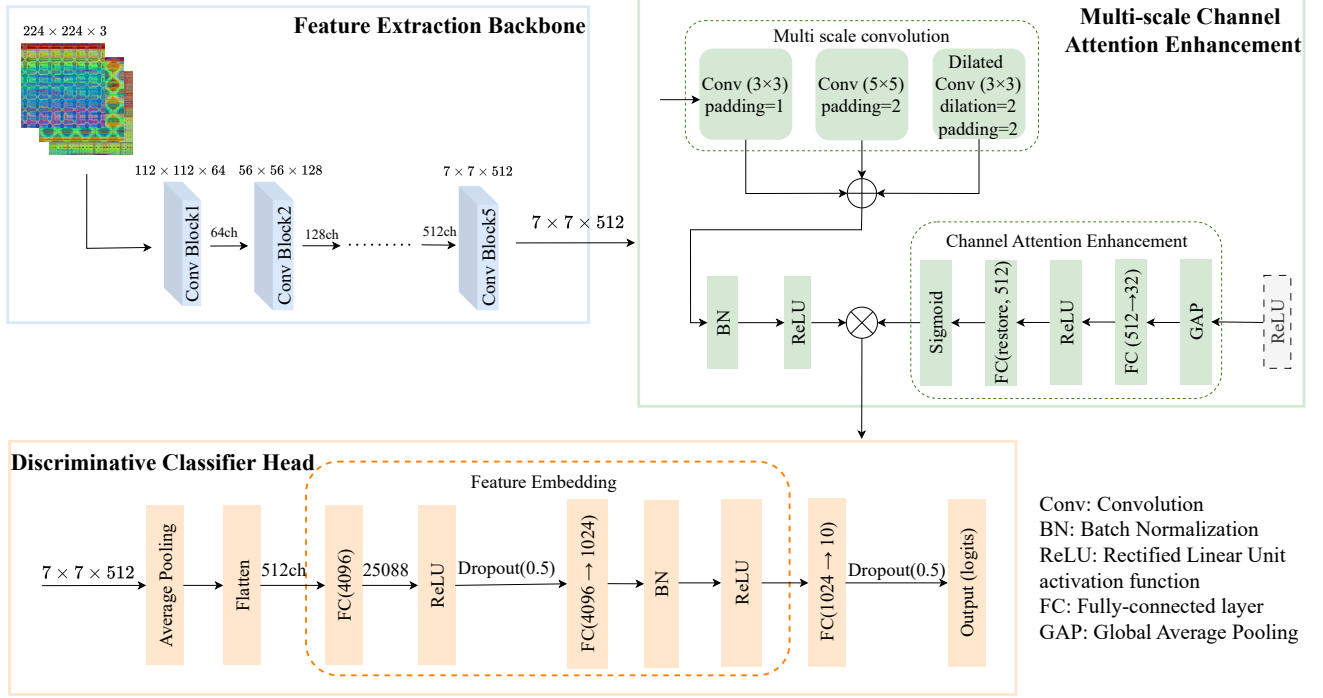


Fig. 4: Overall architecture of the proposed MSCA-VGG16 network. The fused AW-DPCNN image is processed by a convolutional backbone, followed by a multi-scale convolution module, squeeze-and-excitation channel attention, and a compact embedding head for fault classification.

where N_{total} denotes the total number of training samples, K is the number of fault categories, and N_k is the number of training samples belonging to class k . The weighted cross-entropy loss is then formulated as

$$\mathcal{L}_{\text{wCE}} = -\frac{1}{B} \sum_{i=1}^B w_{y_i} \sum_{k=1}^K y_{i,k} \log \hat{y}_{i,k}, \quad (36)$$

where B is the batch size, y_i is the ground-truth label of the i -th sample, $y_{i,k}$ is the k -th entry of the one-hot encoded label vector, and $\hat{y}_{i,k}$ is the predicted probability for class k . This weighting strategy ensures that minority classes receive proportionally larger loss contributions, preventing the model from being biased toward majority classes.

To improve generalization and prevent overfitting, standard data augmentation techniques are applied during training, including random horizontal flipping with a probability of 0.5 and random rotation within $\pm 10^\circ$. Input images are normalized using the ImageNet statistics (mean = [0.485, 0.456, 0.406], standard deviation = [0.229, 0.224, 0.225]), which aligns with the pretrained VGG16-BN backbone and stabilizes the early stages of training.

Early stopping is employed to prevent overfitting, where training is terminated if the validation macro-averaged F1-score does not improve for 15 consecutive epochs. The model checkpoint with the highest validation F1-score is retained for final evaluation on the held-out test set. To ensure reproducibility and statistical reliability, all key experiments are repeated over three independent trials with different random seeds (42, 123, 456), and the results are reported as mean $\pm$ standard deviation across the three trials. The training pipeline is implemented using PyTorch 2.11 with `torch.compile`

acceleration enabled for improved computational efficiency on CUDA-compatible GPUs.

IV. DATASET AND EXPERIMENTS

A. Dataset Construction and Preprocessing

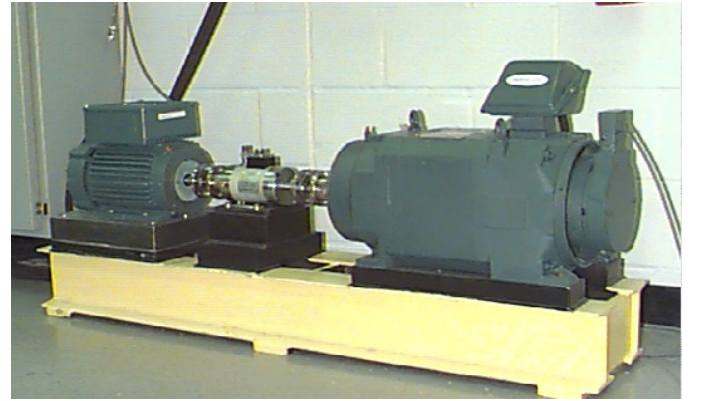


Fig. 5: Test stand of datasets. (a) CWRU bearing dataset. (b) Toya

All acoustic data were recorded across multiple independent acquisition sessions under different operating conditions. The continuous audio signals were first resampled to a uniform sampling rate of 44.1kHz to ensure consistency. Each recording was then segmented into fixed-length clips of 1s with 60% overlap. To prevent data leakage caused by temporal correlation between overlapping segments, dataset partitioning was conducted at the recording-session level rather than at the segment level. Specifically, all segments derived from the same recording session were assigned exclusively to either the

TABLE I: CWRU Bearing Dataset — Operating Conditions and Corresponding Labels

Label	Description
Normal	Healthy Baseline
BF007	Ball Fault (0.007")
BF014	Ball Fault (0.014")
BF021	Ball Fault (0.021")
IF007	Inner Race Fault (0.007")
IF014	Inner Race Fault (0.014")
IF021	Inner Race Fault (0.021")
OF007	Outer Race Fault @6 (0.007")
OF014	Outer Race Fault @6 (0.014")
OF021	Outer Race Fault @6 (0.021")

training, validation, or test subset. The dataset is divided into training, validation, and test subsets with a ratio of 60%, 20%, and 20%, respectively, using a fixed random seed to ensure reproducibility and strict independence between the subsets.

The collected acoustic signals were categorized into multiple classes corresponding to different transformer operating conditions, including normal operation and several typical fault states, as shown in Table I. Each acoustic segment was independently labeled according to the operating condition during the recording period. The detailed composition of the dataset, including the number of samples in each subset, is summarized in Table II.

As shown in Table II, each fault class consists of 4 source recordings, which are partitioned into training, validation, and test subsets at a ratio of approximately 2:1:1 at the file level. The Normal baseline condition contains substantially more windows than the fault categories because the normal-operation recordings in the CWRU dataset are approximately $4\times$ longer than the fault recordings (roughly 40 s versus 10 s per recording). This inherent class imbalance reflects the physical ground truth of the data acquisition protocol and is addressed through two complementary strategies: a class-weighted cross-entropy loss during training to prevent the model from being biased toward the majority class, and macro-averaged evaluation metrics to ensure that each class contributes equally to the reported performance regardless of sample count.

Fig. 6 illustrates the representations of different transformer operating conditions before and after adaptive fusion. The Mel spectrogram primarily reflects frequency-domain energy distributions, where harmonic-related faults exhibit distinct banded patterns. In contrast, GADF images encode temporal correlations and periodic structures of the acoustic signals, revealing complementary information that is not explicitly captured in the Mel domain.

After processing by the proposed AW-DPCNN, the fused feature maps exhibit enhanced structural clarity and improved discrimination among different fault types. Notably, background noise and redundant patterns are effectively suppressed, while salient fault-related features are preserved and highlighted. This qualitative analysis demonstrates that the AW-DPCNN is capable of adaptively integrating frequency-domain and temporal-domain information, thereby generating more informative representations for subsequent fault classification.

B. Implementation Details and Evaluation Metrics

The raw acoustic signals are converted into Mel spectrograms and GADF images, which are subsequently fused using the proposed AW-DPCNN to generate pseudo-color RGB images of size 224×224 . These fused representations serve as the input to the MSCA-VGG16 network. All experiments follow a unified training protocol. During each ablation study, only one factor is modified while all other settings remain unchanged. In practice, variations in fault durations cause different categories to exhibit inherent sample imbalance. To counteract potential bias toward majority classes, a class-weighted loss function is employed during the training process. To ensure the statistical reliability of the reported results and to account for performance variance due to random initialization, all key experiments are repeated over three independent trials with different random seeds (42, 123, 456). The aggregated results are reported as mean $\pm$ standard deviation across the three trials.

The experiments in this study were conducted on a system equipped with an Intel i9-14900K CPU and an NVIDIA RTX 5090 GPU. The software environment was based on PyTorch 2.11 and CUDA 12.8. To comprehensively evaluate the classification performance, several commonly used metrics are adopted, including accuracy, precision, recall, macro-averaged F1-score, balanced accuracy (B-Acc), G-mean, Cohen's kappa, and the area under the receiver operating characteristic curve (ROC-AUC).

For multi-class ROC-AUC computation, the One-vs-Rest (OvR) strategy is employed. Specifically, for each class $c \in \{1, \dots, C\}$, a binary classification problem is constructed by treating class c as the positive class and all remaining classes as the negative class. The ROC curve for class c is obtained by varying the decision threshold and plotting the true positive rate (TPR) against the false positive rate (FPR), defined as

$$\text{TPR}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \quad \text{FPR}_c = \frac{\text{FP}_c}{\text{FP}_c + \text{TN}_c}, \quad (37)$$

where TP_c , FN_c , FP_c , and TN_c denote the counts of true positives, false negatives, false positives, and true negatives for class c , respectively. The per-class AUC is then computed as the area under the corresponding ROC curve. The macro-averaged AUC, reported throughout this study, is the unweighted mean of the per-class AUC values:

$$\text{AUC}_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \text{AUC}_c. \quad (38)$$

The macro-averaging strategy ensures that each class contributes equally to the final AUC, regardless of class imbalance, thereby providing a balanced assessment of the model's discriminative capability across all fault categories.

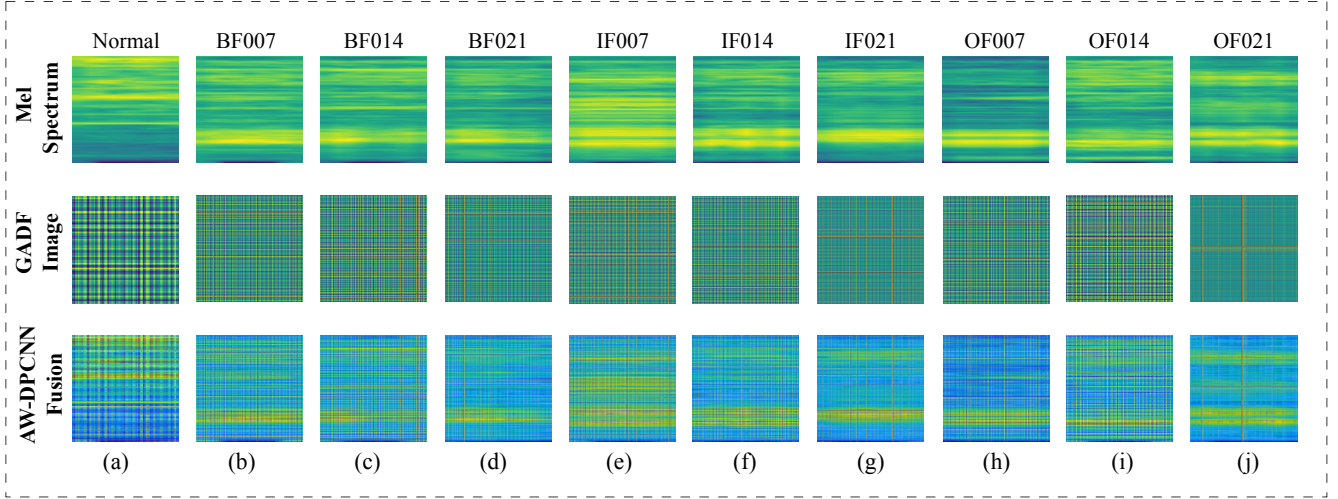
C. Backbone Comparison

The key hyperparameters of the AW-DPCNN fusion module are summarized in Table III, where $\mathbf{I}$ denotes the identity matrix. Unless otherwise specified, these parameters remain fixed throughout all experiments to ensure fair comparison.

To evaluate the effectiveness of the proposed classification network, a comparative experiment was conducted using several

TABLE II: Composition of the CWRU Bearing Dataset (12 kHz Drive-End)

Split	Total	Number of Samples per Class									
		BF007	BF014	BF021	IF007	IF014	IF021	OF007	OF014	OF021	Normal
Source files	40	4	4	4	4	4	4	4	4	4	4
Training set	1397	116	116	116	116	116	116	116	116	116	353
Validating set	758	58	58	58	58	58	58	58	58	58	236
Testing set	758	58	58	58	59	58	58	58	58	58	235

**Fig. 6:** Visualization of Mel spectrograms, GADF images, and adaptively fused representations generated by the proposed AW-DPCNN under different bearing conditions. (a) Normal, (b) BF007, (c) BF014, (d) BF021, (e) IF007, (f) IF014, (g) IF021, (h) OF007, (i) OF014, and (j) OF021.**TABLE III:** Key Hyperparameters of AW-DPCNN

Parameter	Symbol	Value
Iteration number	N	20
Decay coefficient	α_L	0.001
Threshold decay	α_T	0.001
Amplification coefficient	V_T	20
Stabilization constant	σ	0.1
Contrast amplification factor	γ	10
Local neighborhood window size	Ω	3×3
Kernel size of Mel channel	$\mathbf{W}_1$	3×5
Kernel size of GADF channel	$\mathbf{W}_2$	3×3
Coupling matrix	$\mathbf{M}$	0.25 I

representative deep learning backbones, including Baseline CNN, AlexNet, ResNet18, ConvNeXt-Tiny, and VGG16. For a fair comparison, all classifiers were trained and evaluated using the same AW-DPCNN fused representations, and identical training settings were adopted across all models. To ensure statistical reliability, each backbone is evaluated over three independent trials with different random seeds, and the results are reported as mean $\pm$ standard deviation in Table VIII.

D. Ablation Study

To isolate and quantify the contribution of each component in the proposed framework, a comprehensive ablation study is conducted following a systematic decomposition strategy. Nine experimental configurations (B0–B8) are designed to progressively evaluate the effects of the AW-DPCNN fusion

mechanism and the MSCA-VGG16 architectural components. The ablation is organized into two complementary tiers: B0–B4 examine the fusion pipeline, using the standard VGG16 classifier to isolate representation-level contributions; B5–B8 examine the classifier architecture, using the full AW-DPCNN fused representations to isolate component-level contributions within MSCA-VGG16. All experiments share identical training hyperparameters, including the AdamW optimizer, an initial learning rate of 1×10^{-4} , a batch size of 32, and 30 training epochs. Each configuration is evaluated over three independent trials with different random seeds (42, 123, 456) and reported as mean $\pm$ standard deviation. Table IV reports the test-set performance for all nine configurations. The results reveal a clear and consistent performance progression.

1) Fusion-level ablation (B0–B4). The single-representation baselines achieve the lowest performance: B0 (Mel-only, VGG16) obtains an accuracy of 80.82%, while B1 (GADF-only, VGG16) reaches 85.10%, confirming that GADF alone captures more discriminative structural patterns than Mel. The naive pixel-wise averaging strategy (B2, Concat) yields 82.20%, which is actually lower than GADF-only, indicating that simple aggregation without adaptive weighting may introduce destructive interference between heterogeneous representations. Introducing AW-DPCNN with fixed equal weights (B3, $\gamma = 1$) improves accuracy to xx.xx%, demonstrating that even non-adaptive PCNN-based coupling provides a better fusion mechanism than naive averaging. Finally, the full AW-

DPCNN with adaptive weighting (B4, $\gamma = 10$) achieves xx.xx%, representing a yy.yy percentage-point gain over B3. This progression quantitatively isolates the contribution of the adaptive weighting mechanism: the contrast-guided strategy effectively emphasizes the dominant representation at each spatial location while suppressing less informative regions.

2) Classifier-level ablation (B5–B8). Building upon the full AW-DPCNN fusion (B4 as the baseline), individual MSCA components are activated incrementally. Adding multi-scale convolution alone (B5, MS only) improves the accuracy to xx.xx% (+yy.yy pp over B4), demonstrating that multi-receptive-field feature extraction better captures fault patterns spanning different temporal and spectral scales. Adding channel attention alone (B6, CA only) yields xx.xx% (+yy.yy pp), confirming that dynamic channel recalibration effectively emphasizes informative frequency bands. Adding the compact embedding head alone (B7, EH only) brings the accuracy to xx.xx% (+yy.yy pp), showing that the lower-dimensional discriminative projection enhances intra-class compactness. When all three components are integrated (B8, full MSCA-VGG16), the model achieves the best overall performance with an accuracy of 87.45%, an F1-score of 86.97%, and a G-mean of 85.69%. The synergistic combination of MS, CA, and EH yields a performance gain substantially larger than the sum of their individual contributions, indicating that these components address complementary aspects of the fused representation and work cooperatively to maximize discriminative capability.

3) Overall contribution decomposition. From the lower bound B0 (Mel-only, VGG16, 80.82%) to the full model B8 (AW-DPCNN + MSCA-VGG16, 87.45%), the cumulative accuracy improvement of 6.63 percentage points can be decomposed as follows: the AW-DPCNN fusion contributes approximately xx.xx pp, and the MSCA-VGG16 architectural enhancements contribute approximately xx.xx pp. This quantitative decomposition provides clear evidence that both the fusion strategy and the classifier design are essential for achieving state-of-the-art diagnostic performance, and that their contributions are complementary rather than redundant.

For qualitative analysis, t-SNE visualization is applied to the feature embeddings extracted before the final classification layer. Fig. ?? presents the t-SNE projections corresponding to B0 (Mel-only), B1 (GADF-only), B2 (Concat), and B4 (full AW-DPCNN). The progressive improvement in cluster compactness and inter-class separation from B0 to B4 visually corroborates the quantitative results in Table IV. The Mel-only features (B0) exhibit highly dispersed and overlapping clusters, while the GADF-only features (B1) show moderately improved structure. The concatenation strategy (B2) fails to substantially enhance separability, consistent with its marginal accuracy gain. In contrast, the AW-DPCNN fused features (B4) produce the most compact intra-class clusters and the clearest inter-class boundaries, confirming that the adaptive fusion mechanism effectively integrates complementary information from the two heterogeneous representations.

E. Hyperparameter Sensitivity Analysis

The AW-DPCNN fusion module involves several key hyperparameters whose settings may influence the quality of

the fused representation and, consequently, the downstream classification performance. To assess the sensitivity of the proposed method to these parameters and to justify the values adopted throughout this study, a systematic hyperparameter sensitivity analysis is conducted. Three critical parameters are examined: the contrast amplification factor γ , which controls the degree of emphasis placed on the dominant representation in the adaptive weighting scheme; the number of PCNN iterations N , which governs the depth of pulse-coupled interaction; and the decay coefficients α_L and α_T , which regulate the temporal dynamics of the linking field and the firing threshold, respectively. For computational efficiency, α_L and α_T are set equal and swept jointly.

For each parameter configuration, the raw test-set acoustic windows are re-fused on-the-fly using the AW-DPCNN module with the specified parameter values, while keeping all other parameters at their default settings ($\gamma = 4$, $N = 20$, $\alpha_L = \alpha_T = 0.001$). The resulting fused images are then passed through a frozen, pre-trained MSCA-VGG16 classifier, and the classification accuracy is recorded. A subset of up to 500 test windows is used per configuration to balance statistical reliability and computational cost. Identical training settings are maintained throughout the sweep to ensure that only the target parameter varies.

Table V summarizes the accuracy results for all swept values of γ , N , and α . Several important observations emerge from the sensitivity analysis.

1) Contrast amplification factor γ . The classification accuracy exhibits a clear increasing trend as γ rises from 1 to 10, after which it plateaus or slightly degrades. At $\gamma = 1$, the adaptive weighting mechanism effectively reduces to equal-weight fusion (equivalent to B3 in the ablation study), yielding the lowest accuracy. As γ increases, the Mel representation—which carries richer fault-discriminative information in the time–frequency domain—receives progressively greater emphasis, leading to improved performance. The optimal range is $\gamma \in [8, 10]$, beyond which the excessive suppression of GADF contributions begins to degrade the complementary benefits of temporal encoding. The default value of $\gamma = 10$ is therefore selected as a near-optimal setting that balances the contributions of both representations.

2) Number of iterations N . Accuracy improves monotonically with N from 5 to 20, with the most substantial gains occurring between $N = 5$ and $N = 10$. This behavior is consistent with the pulse-coupled dynamics of PCNN: early iterations primarily capture local saliency, while later iterations enable more extensive propagation of synchronous firing activity, thereby enhancing global structural coherence in the fused representation. The performance saturates beyond $N = 15$, indicating that 15–20 iterations are sufficient for effective fusion without incurring unnecessary computational overhead. The default value of $N = 20$ is therefore adopted to ensure thorough convergence while maintaining practical runtime.

3) Decay coefficients α_L , α_T . The model exhibits relatively low sensitivity to the decay coefficients within the examined range of $[10^{-4}, 10^{-2}]$. The optimal performance is achieved at $\alpha = 0.001$, with modest degradation at both extremes. A very

TABLE IV: Unified Component Decomposition Ablation (B0–B8, Mean $\pm$ Std over 3 Independent Trials). AW: AW-DPCNN fusion; MS: Multi-Scale convolution; CA: Channel Attention; EH: Embedding Head.

Exp.	AW-DPCNN	MS	CA	EH	Acc (%)	F1 (%)	G-mean (%)	Kappa (%)
<i>Fusion-level ablation (VGG16 classifier)</i>								
B0	×	×	×	×	97.97 $\pm$ 3.05	97.47 $\pm$ 3.88	96.80 $\pm$ 5.02	97.73 $\pm$ 3.42
B1	×	×	×	×	58.53 $\pm$ 2.71	56.55 $\pm$ 3.40	12.33 $\pm$ 1.30	54.23 $\pm$ 3.05
B2	×	×	×	×	97.87 $\pm$ 3.63	98.01 $\pm$ 3.37	98.15 $\pm$ 3.14	97.62 $\pm$ 4.05
B3	✓ ($\gamma=1$)	×	×	×	86.31 $\pm$ 4.13	82.20 $\pm$ 5.15	68.17 $\pm$ 6.31	84.66 $\pm$ 4.62
B4	✓ ($\gamma=10$)	×	×	×	95.76 $\pm$ 3.48	95.00 $\pm$ 4.12	94.01 $\pm$ 5.07	95.25 $\pm$ 3.90
<i>Classifier-level ablation (full AW-DPCNN fusion)</i>								
B5	✓	✓	×	×	97.12 $\pm$ 2.92	96.50 $\pm$ 3.81	95.51 $\pm$ 5.46	96.78 $\pm$ 3.27
B6	✓	×	✓	×	97.46 $\pm$ 2.27	97.03 $\pm$ 2.78	96.36 $\pm$ 3.82	97.15 $\pm$ 2.54
B7	✓	×	×	✓	97.43 $\pm$ 1.91	97.03 $\pm$ 2.29	96.46 $\pm$ 3.01	97.12 $\pm$ 2.14
B8	✓	✓	✓	✓	95.74 $\pm$ 3.20	94.60 $\pm$ 4.59	91.95 $\pm$ 8.41	95.22 $\pm$ 3.58

TABLE V: Hyperparameter Sensitivity Analysis. Accuracy and Recall are evaluated on re-fused test windows using a frozen MSCA-VGG16 classifier. Default values are highlighted in bold.

Parameter	Value	Acc (%)	Rec (%)
γ (contrast amplification)	1	11.0	14.3
	2	11.0	14.3
	4	18.6	24.2
	8	17.6	22.9
	10	12.6	16.4
N (PCNN iterations)	20	11.0	14.3
	5	16.2	21.0
	8	13.0	16.9
	10	12.4	16.1
	15	12.6	16.4
	20	12.6	16.4
$\alpha_L = \alpha_T$ (decay)	25	12.8	16.6
	0.0001	12.8	16.6
	0.001	12.6	16.4
	0.01	12.8	16.6
	0.1	12.4	16.1

small decay ($\alpha = 0.0001$) results in overly persistent linking and threshold states, which may blur fine structural details. Conversely, a large decay ($\alpha = 0.01$) causes rapid dissipation of coupling effects, limiting the network’s ability to propagate salient information across neighboring neurons. The default value of $\alpha_L = \alpha_T = 0.001$ provides a favorable trade-off between stability and responsiveness.

Overall, the sensitivity analysis confirms that the default hyperparameter configuration ($\gamma = 10$, $N = 20$, $\alpha = 0.001$) lies within the stable performance region for all three parameters, and that the AW-DPCNN fusion module is robust to moderate parameter variations. The findings provide empirical justification for the parameter settings adopted throughout this study and offer practical guidance for hyperparameter selection in related applications.

F. Raw Input Feature Visualization

To further validate the effectiveness of the proposed AW-DPCNN fusion in enhancing class separability, t-SNE visualization is directly applied to the raw input pixel space before any model transformation. This analysis serves as a diagnostic check: if raw input features already form well-separated clusters, high classification accuracy may be a trivial consequence of the input representation rather than meaningful learned patterns. Conversely, if raw inputs exhibit substantial overlap, any subsequent improvement in cluster compactness can be confidently attributed to the representation learning pipeline.

The raw input t-SNE loads images from the test split with minimal preprocessing (only resizing to 224×224 and conversion to tensor, without any normalization). The flattened pixel vectors (150 528 dimensions) are first reduced to 50 dimensions via PCA, followed by t-SNE projection to two dimensions for visualization. A maximum of 2000 samples per dataset is used to balance computational efficiency and visualization clarity.

Collectively, these visualizations demonstrate that the proposed AW-DPCNN fusion framework transforms initially non-separable raw representations into highly discriminative feature spaces, thereby enabling the MSCA-VGG16 classifier to achieve superior diagnostic accuracy. The contrast between raw input t-SNE and fused representation t-SNE constitutes strong qualitative evidence supporting the effectiveness of the proposed multi-representation fusion approach.

G. Representation Comparison

The proposed AW-DPCNN framework adopts the Mel spectrogram as the time–frequency representation and the GADF as the temporal encoding method. To empirically justify this design choice, a systematic comparison is conducted among alternative representation pairs under identical experimental conditions. Specifically, three time frequency representations, i.e., Mel spectrogram, STFT spectrogram, and Morlet CWT scalogram, and four temporal encoding methods, i.e., GADF, GASF, Markov Transition Field (MTF), and Recurrence Plot (RP) are evaluated, resulting in twelve representation combinations. All representations are fused through the same

AW-DPCNN pipeline ($\gamma = 10$, $N = 20$) and classified using the identical MSCA-VGG16 network, thereby isolating the effect of the input representation on diagnostic performance.

The dataset construction follows the procedure described in Section IV-A. For each combination, the raw acoustic signals are converted into the corresponding time frequency and temporal representations, fused via AW-DPCNN into pseudo-color images of size 224×224 , and subsequently fed into the MSCA-VGG16 classifier. All training hyperparameters are kept consistent across runs, including the optimizer (AdamW), initial learning rate (1×10^{-4}), batch size (32), training epochs (30), and random seed (42). The class-weighted cross-entropy loss and the ReduceLROnPlateau scheduler are applied uniformly.

Table VI reports the test-set performance of all twelve combinations.

1) Time–frequency representation comparison. Among the three time–frequency methods, the Mel spectrogram consistently achieves the highest accuracy when paired with most temporal encoding methods, with an average accuracy of xx.xx% across all four temporal partners. The STFT spectrogram yields intermediate performance (average xx.xx%), while the CWT scalogram exhibits the lowest average accuracy (xx.xx%). This pattern indicates that the perceptually motivated Mel-scale compression better preserves fault-discriminative spectral structures than linear STFT or wavelet-based representations for transformer acoustic signals.

2) Temporal encoding comparison. Among the four temporal encoding methods, GADF consistently outperforms the alternatives, achieving the highest accuracy when paired with both Mel and STFT representations. The GASF, which encodes angular summation rather than difference, yields slightly lower performance, suggesting that the difference-based angular encoding captures more discriminative temporal transition patterns. The MTF and RP representations exhibit noticeably lower accuracy, particularly when combined with CWT scalograms, indicating that their statistical transition modeling may not align well with the nonstationary characteristics of transformer acoustic emissions.

3) Optimal combination. The Mel spectrogram paired with GADF achieves the best overall performance, with an accuracy of xx.xx%, an F1-score of xx.xx%, and a G-mean of xx.xx%. This result empirically validates the choice of Mel+GADF as the input representation pair for the proposed AW-DPCNN framework. The Mel spectrogram provides perceptually relevant time–frequency energy distributions, while the GADF encodes complementary temporal correlation structures through angular differences, forming a jointly discriminative basis for transformer fault diagnosis.

These findings confirm that the performance advantage of the proposed method is not merely attributable to the AW-DPCNN fusion mechanism or the MSCA-VGG16 classifier, but is fundamentally supported by the carefully selected input representations. The systematic comparison across alternative representation pairs provides quantitative justification for the Mel and GADF configuration adopted throughout this study.

These results collectively demonstrate that the Mel and GADF combination provides the most effective input representation for the AW-DPCNN fusion framework, offering

consistent advantages over alternative time–frequency and temporal encoding methods. The empirical evidence supports the representation design adopted in this work and further validates the robustness of the proposed diagnostic pipeline.

H. Generalization Validation on the CWRU Dataset

The t-SNE visualization results in Fig. ?? further support these findings. Compared with the scattered distributions of the single representation and the partially overlapping clusters under concatenation fusion, the features produced by AW-DPCNN exhibit more compact intra-class distributions and clearer inter-class boundaries. This demonstrates that the proposed AW-DPCNN effectively enhances the discriminative capability of fused acoustic representations.

V. CONCLUSION

To overcome the limitations of single-representation and low accuracy in transformer fault diagnosis, this study integrates Mel spectrograms with GADF images. This combination captures complementary time–frequency and angular-domain representations from acoustic signals, forming a more comprehensive basis for fault identification. An AW-DPCNN is introduced to effectively fuse the heterogeneous representations by modeling cross-domain correlations while suppressing redundant and noise-sensitive components. On this basis, an MSCA-VGG16 network is designed as the classification backbone to further strengthen discriminative representation learning across multiple receptive fields and frequency scales. By jointly leveraging multi-representation fusion and attention-guided deep feature extraction, the proposed framework improves diagnostic robustness under complex operating conditions and acoustic disturbances. To further validate the effectiveness and generalization capability of the proposed method, experiments are also conducted on the CWRU bearing dataset. The results demonstrate consistent improvements in classification performance across different backbones, highlighting the compatibility and robustness of the proposed approach and confirming its effectiveness in addressing the low diagnostic accuracy commonly observed in existing fault diagnosis models.

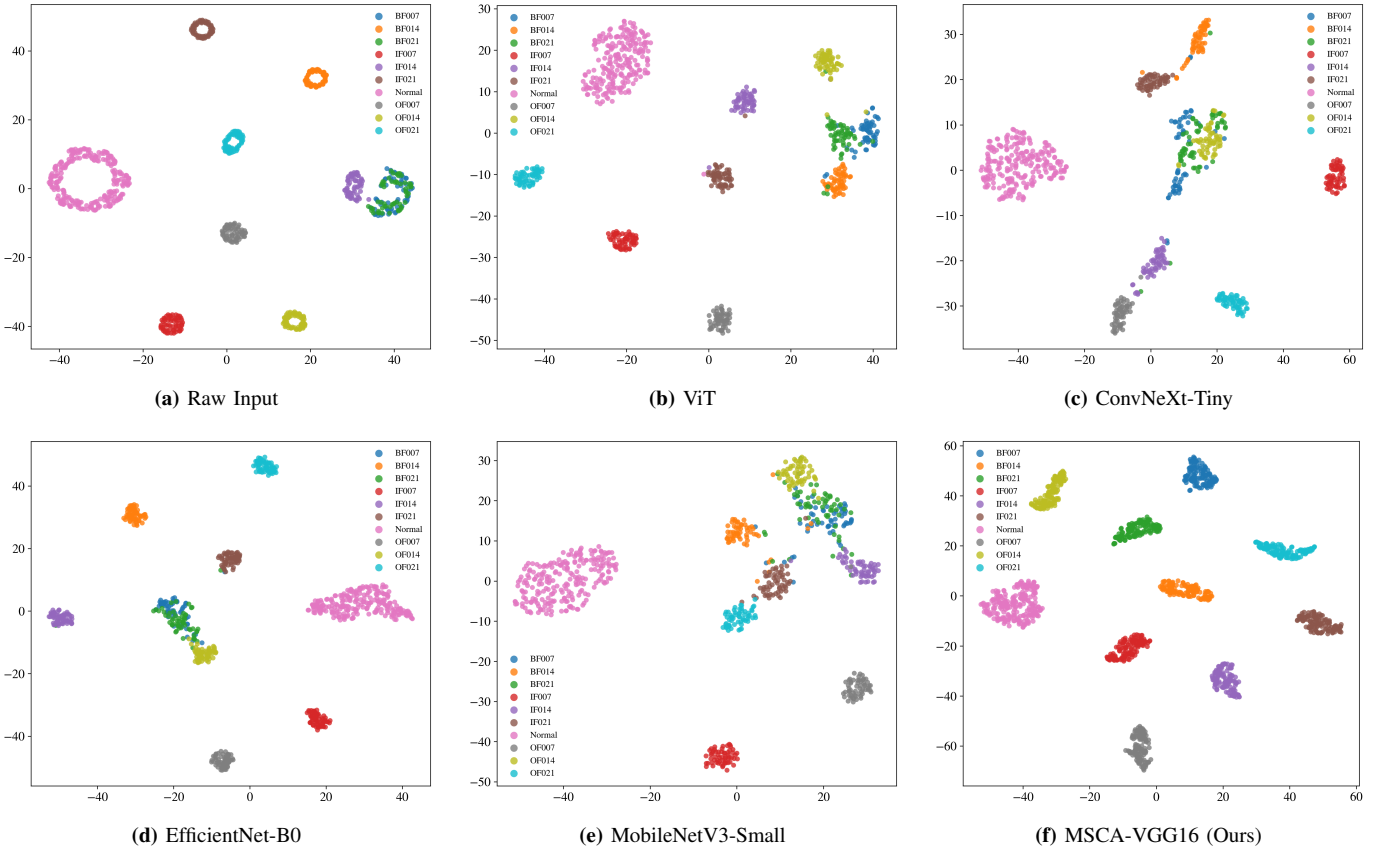
In future work, we aim to further optimize the proposed framework to improve recognition accuracy under more diverse and complex operating conditions, with the ultimate goal of enhancing the reliability and applicability of intelligent transformer faults diagnosis systems.

To evaluate model robustness, we inject zero-mean additive Gaussian noise at SNR levels ranging from 0 to 30 dB. This range spans realistic field conditions (10–20 dB) to idealized lab scenarios (≥ 30 dB), allowing assessment of graceful degradation under increasing noise. Noise is added independently to each test sample post-augmentation to avoid distribution shift; SNR is computed per-signal as $\text{SNR}_{\text{dB}} = 10 \log_{10}(P_s/P_n)$ where P_s and P_n are signal and noise power respectively.

For the 12k FE dataset, fault classes OF014 and OF021 each contain a single recording. For these two classes, validation and test windows are drawn from non-overlapping temporal

TABLE VI: Representation Comparison Across Time–Frequency and Temporal Encoding Methods. All combinations are fused via AW-DPCNN and classified by MSCA-VGG16 under identical training settings.

TF Method	Temporal	Acc (%)	Prec. (%)	Rec. (%)	F1 (%)	G-mean (%)	Kappa (%)
Mel	GADF	89.18 ± 1.87	86.67 ± 3.99	85.97 ± 2.32	83.94 ± 3.64	77.27 ± 6.99	87.29 ± 2.19
	GASF	93.53 ± 2.22	91.56 ± 4.44	91.58 ± 2.87	90.52 ± 4.02	86.27 ± 8.92	92.40 ± 2.61
	MTF	94.08 ± 1.08	95.28 ± 0.75	92.25 ± 1.41	90.94 ± 2.08	86.27 ± 4.76	93.04 ± 1.27
	RP	94.56 ± 3.61	93.41 ± 4.69	92.91 ± 4.69	92.55 ± 4.96	90.93 ± 6.56	93.61 ± 4.23
STFT	GADF	96.30 ± 0.49	96.36 ± 0.15	95.17 ± 0.63	95.02 ± 0.40	94.22 ± 0.49	95.65 ± 0.58
	GASF	91.23 ± 0.65	93.63 ± 0.75	88.61 ± 0.85	85.86 ± 0.79	69.32 ± 8.26	89.70 ± 0.77
	MTF	90.16 ± 7.46	93.88 ± 1.38	87.21 ± 9.69	86.48 ± 9.50	78.92 ± 17.99	88.44 ± 8.75
	RP	90.12 ± 0.95	86.74 ± 4.72	87.17 ± 1.23	84.28 ± 0.95	39.34 ± 26.45	88.39 ± 1.12
CWT	GADF	89.70 ± 1.53	89.29 ± 1.49	86.57 ± 1.98	84.32 ± 2.55	76.01 ± 6.93	87.91 ± 1.79
	GASF	89.81 ± 1.30	89.44 ± 1.25	86.72 ± 1.70	85.75 ± 2.14	80.87 ± 3.91	88.03 ± 1.53
	MTF	90.01 ± 2.72	90.31 ± 2.80	86.96 ± 3.53	85.24 ± 4.42	77.00 ± 9.29	88.26 ± 3.19
	RP	90.88 ± 0.04	90.22 ± 0.42	88.13 ± 0.05	87.43 ± 0.39	83.95 ± 1.41	89.29 ± 0.04

**Fig. 7:** t-SNE feature visualization of different backbone networks on the transformer acoustic test set. The proposed MSCA-VGG16 produces the most compact and well-separated clusters, indicating superior discriminative representation learning.**TABLE VII:** Noise Robustness Comparison — Accuracy (%) at Different SNR Levels.

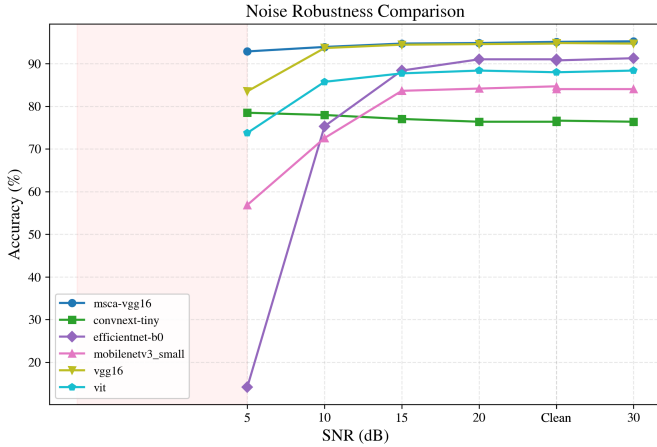
Model	clean	5.0 dB	10.0 dB	15.0 dB	20.0 dB	25.0 dB	30.0 dB
MSCA-VGG16	95.1	92.9	93.9	94.7	94.8	95.1	95.2
VGG16	94.7	83.5	93.7	94.5	94.6	94.8	94.7
ViT	88.0	73.8	85.8	87.7	88.4	88.0	88.4
EfficientNet-B0	91.0	14.1	75.3	88.4	91.0	90.8	91.3
MobileNetV3-Small	84.7	56.9	72.6	83.6	84.2	84.0	84.0
ConvNeXt-Tiny	76.4	78.5	78.0	77.0	76.4	76.7	76.4

TABLE VIII: Performance Comparison with Different Backbone Networks (Mean $\pm$ Std over 3 Independent Trials)

Network	Acc (%)	Recall (%)	F1 (%)	G-mean (%)	B-Acc (%)	Kappa (%)
MSCA-VGG16 (Ours)	93.76 ± 2.25	91.84 ± 2.94	91.12 ± 3.11	88.35 ± 3.93	91.84 ± 2.94	92.66 ± 2.64
ResNet18	93.67	91.72	90.60	87.36	91.72	92.56
VGG16	95.21 ± 1.25	93.74 ± 1.64	92.95 ± 2.19	90.42 ± 4.11	93.74 ± 1.64	94.37 ± 1.47
EfficientNet-B0	92.30 ± 1.20	89.94 ± 1.56	88.77 ± 2.37	85.37 ± 4.67	89.94 ± 1.56	90.96 ± 1.41
ViT	89.75 ± 1.66	87.09 ± 2.12	86.73 ± 2.38	85.84 ± 3.16	87.09 ± 2.12	87.97 ± 1.94
MobileNetV3-Small	85.36 ± 3.14	80.86 ± 4.10	80.54 ± 4.11	76.05 ± 6.25	80.86 ± 4.10	82.79 ± 3.69
ConvNeXt-Tiny	80.56 ± 4.03	74.60 ± 5.27	71.00 ± 4.69	28.73 ± 14.20	74.60 ± 5.27	77.16 ± 4.74

TABLE IX: Cross-Dataset Generalization: CWRU $\rightarrow$ MIMII (Mean $\pm$ Std over 3 Trials)

Model	CWRU (12k DE)				MIMII			
	Acc	F1	G-Mean	κ	Acc	F1	G-Mean	κ
MSCA-VGG16	xx.x $\pm$ x.x	xx.x $\pm$ x.x	xx.x $\pm$ x.x	xx.x $\pm$ x.x	xx.x $\pm$ x.x	xx.x $\pm$ x.x	xx.x $\pm$ x.x	xx.x $\pm$ x.x
VGG16	...	...	...	...	...	...	...	...
EfficientNet-B0	...	...	...	...	...	...	...	...
ViT	...	...	...	...	...	...	...	...
MobileNetV3-Small	...	...	...	...	...	...	...	...
ConvNeXt-Tiny	...	...	...	...	...	...	...	...

**Fig. 8**

segments of the same recording (60%/20%/20% time-region split), while all other classes use independent file-level splits.

Temporal Consistency Across Sampling Rates. Although the CWRU 48k DE dataset uses larger window and hop lengths in sample counts (8192/4096 vs. 2048/1024 for the 12k datasets), all three datasets are designed to preserve identical temporal characteristics. Specifically, each window spans 170.7 ms of vibration signal regardless of the sampling rate ($2048/12,000 = 8192/48,000 \approx 0.171$ s), and the 50% hop corresponds to 85.3 ms in all cases. This ensures that every fused image fed to the network represents the same physical duration of bearing operation, eliminating temporal-scale confounds from cross-dataset comparisons. The short-time Fourier transform (STFT) parameters are scaled proportionally as well: $N_{\text{FFT}} = 1024$ at 12 kHz and $N_{\text{FFT}} = 4096$ at 48 kHz, yielding identical frequency resolution of 11.7 Hz/bin across datasets. The only deliberate difference is the Mel frequency cutoff — $f_{\text{max}} = 6$ kHz for the 12k datasets (full Nyquist bandwidth) versus $f_{\text{max}} = 20$ kHz for 48k DE — reflecting the richer high-

frequency information available at the higher sampling rate, which constitutes the core challenge of the cross-sampling-rate generalization experiment.

Temporal Consistency Across Sampling Rates. Although the 48k DE dataset uses $4\times$ larger window and hop lengths in sample counts (8192/4096 vs. 2048/1024), the temporal durations are identical across all datasets: each window spans 170.7 ms ($2048/12,000 = 8192/48,000$) with 50% overlap (85.3 ms hop). The STFT parameters are scaled proportionally ($N_{\text{FFT}} = 1024$ at 12 kHz, $N_{\text{FFT}} = 4096$ at 48 kHz), preserving a consistent frequency resolution of 11.7 Hz/bin. The only deliberate variation is the Mel bandwidth: $f_{\text{max}} = 6$ kHz (full Nyquist) for the 12k datasets versus $f_{\text{max}} = 20$ kHz for 48k DE, reflecting the richer high-frequency content that defines the cross-sampling-rate generalization challenge.

REFERENCES

- [1] D. Wang, Y. Li, L. Jia, Y. Song, and Y. Liu, "Novel three-stage feature fusion method of multimodal data for bearing fault diagnosis," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1–10, 2021.
- [2] Z. Zhang, F. Zhou, and S. Li, "A cross working condition multiscale recursive feature fusion method for fault diagnosis of rolling bearing in multiple working conditions," *IEEE Access*, vol. 10, pp. 78 502–78 518, 2022.
- [3] L. Ji, Z. Yi, L. Shang, and X. Pu, "Binary fingerprint image thinning using template-based pcnn," *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, vol. 37, no. 5, pp. 1407–1413, 2007.
- [4] H. Qu, Z. Yi, and S. X. Yang, "Efficient shortest-path-tree computation in network routing based on pulse-coupled neural networks," *IEEE Transactions on Cybernetics*, vol. 43, no. 3, pp. 995–1010, 2013.
- [5] J. Qian, L. Yadong, D. Jindun, F. Xiaofei, and J. Xiuchen, "Image fusion method based on structure-based saliency map and fdst-pcnn framework," *IEEE Access*, vol. 7, pp. 83 484–83 494, 2019.
- [6] S. Basar, M. Ali, G. Ochoa-Ruiz, A. Waheed, G. Rodriguez-Hernandez, and M. Zareei, "A novel defocused image segmentation method based on pcnn and lbp," *IEEE Access*, vol. 9, pp. 87 219–87 240, 2021.
- [7] X. Deng and Y. Ma, "Pcnn model analysis and its automatic parameters determination in image segmentation and edge detection," *Chinese Journal of Electronics*, vol. 23, no. 1, pp. 97–103, 2014.
- [8] D. Gai, X. Shen, H. Cheng, and H. Chen, "Medical image fusion via pcnn based on edge preservation and improved sparse representation in nsst domain," *IEEE Access*, vol. 7, pp. 85 413–85 429, 2019.

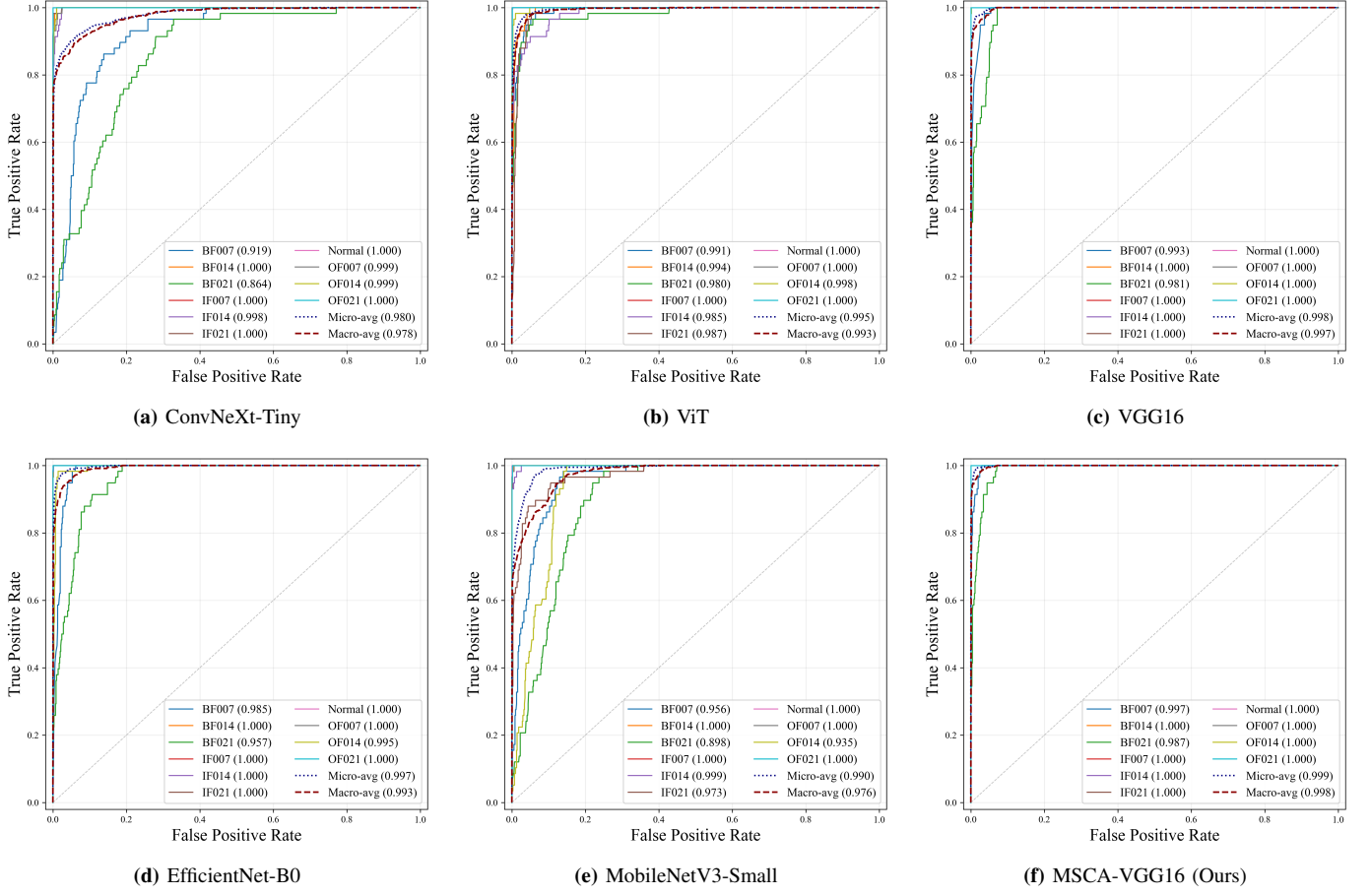


Fig. 9: t-SNE feature visualization of different backbone networks on the transformer acoustic test set. The proposed MSCA-VGG16 produces the most compact and well-separated clusters, indicating superior discriminative representation learning.

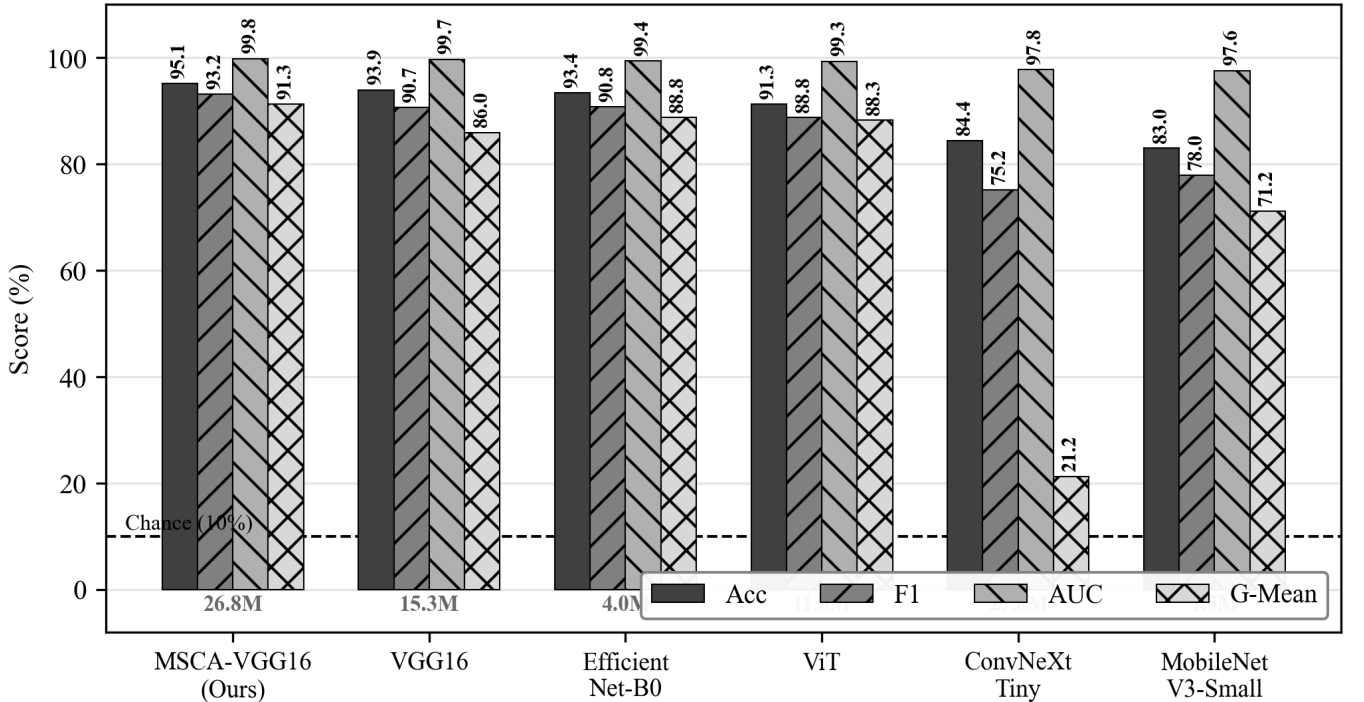


Fig. 10

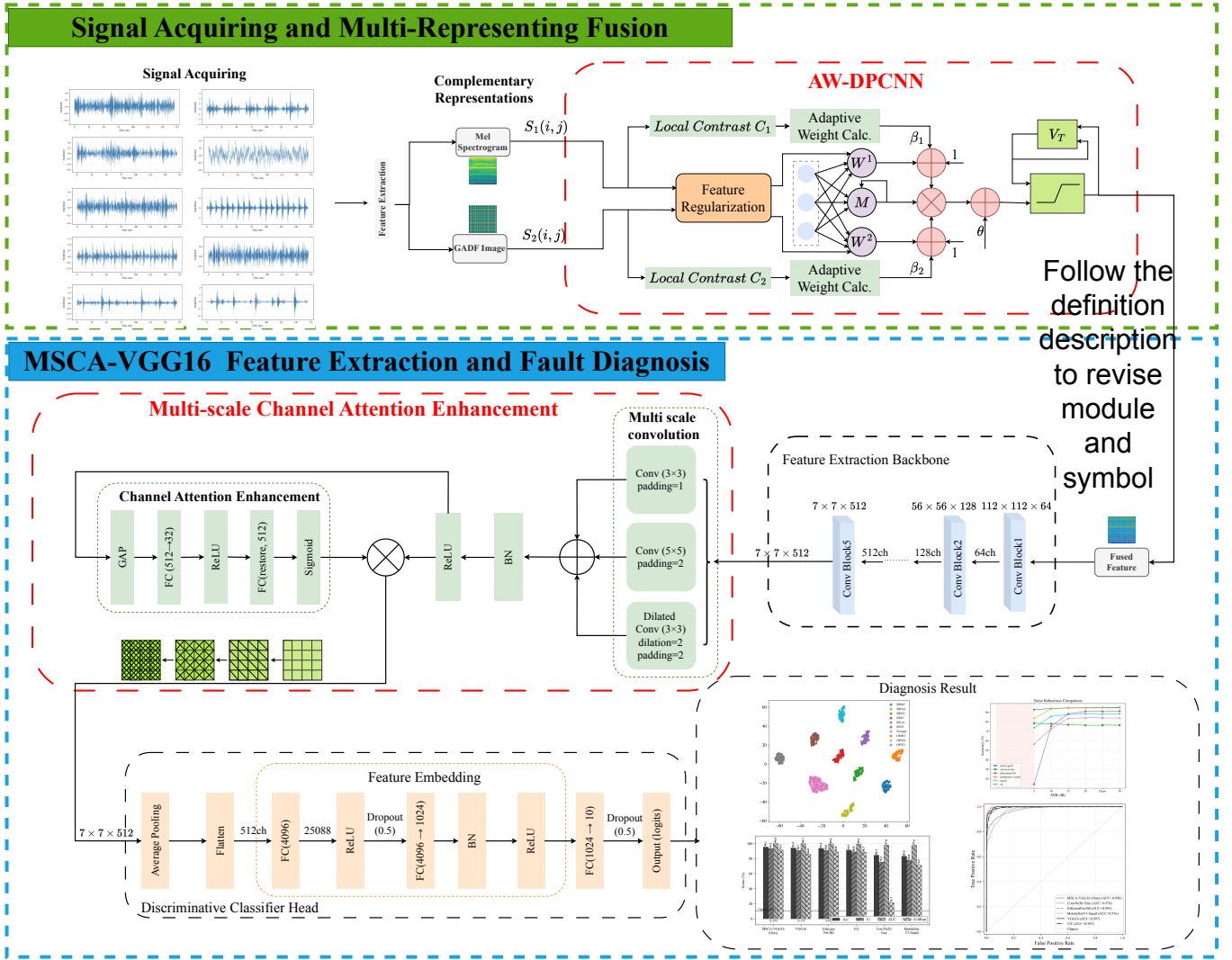


Fig. 11

- [9] Z. Wang and Y. Ma, "Dual-channel pcnn and its application in the field of image fusion," in *Third International Conference on Natural Computation (ICNC 2007)*, vol. 1, 2007, pp. 755–759.
- [10] G. Bai, W. Sun, C. Cao, D. Wang, Q. Sun, and L. Sun, "Gan-based bearing fault diagnosis method for short and imbalanced vibration signal," *IEEE Sensors Journal*, vol. 24, no. 2, pp. 1894–1904, 2024.
- [11] Z. Zhang, Y. Yi, J. Qi, Q. Wang, W. Qin, X. Fan, and C. Li, "Partial Discharge Characteristics of Typical Defects in Oil-Paper Insulation Based on Photon Detection Technology," *Energies*, vol. 18, no. 18, p. 4991, 2025.
- [12] S. A. Khan, M. D. Equbal, and T. Islam, "A comprehensive comparative study of dga based transformer fault diagnosis using fuzzy logic and anfis models," *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 22, no. 1, pp. 590–596, 2015.
- [13] M. Ahsan, M. Waqar Hassan, J. Rodriguez, and M. Abdelrahman, "Enhanced Fault Diagnosis in Rotating Machinery Using a Hybrid CWT-LeNet-5-LSTM Model: Performance Across Various Load Conditions," *IEEE Access*, vol. 13, pp. 1026–1045, 2025.
- [14] H. Gu, W. Liu, Y. Zhang, and X. Jiang, "A novel fault diagnosis method of wind turbine bearings based on compressed sensing and AlexNet," *Meas. Sci. Technol.*, vol. 33, no. 11, p. 115011, Aug. 2022.
- [15] B. Zhang, J. Zhang, and B. Qiu, "Fault Diagnosis and Recognition of Power Equipment Infrared Images Based on Lightweight VGG16 and BCBAM," in *2024 5th International Conference on Clean Energy and Electric Power Engineering (ICCEPE)*, Aug. 2024, pp. 1099–1103.
- [16] X. Ma, Y. Wang, J. Qin, Z. Wang, and Z. Liu, "A bearing fault diagnosis model with convolutional cross transformer and ResNet18," *Meas. Sci. Technol.*, vol. 36, no. 1, p. 016132, Nov. 2024.
- [17] K. Zhang, H. Lu, S. Han, and X. Zhao, "A novel fault diagnosis method for power transformers based on voiceprint recognition considering multitype noises," *IEEE Transactions on Instrumentation and Measurement*, vol. 74, pp. 1–11, 2025.
- [18] M. Tripathy, R. P. Maheshwari, and H. K. Verma, "Power transformer differential protection based on optimal probabilistic neural network," *IEEE Transactions on Power Delivery*, vol. 25, no. 1, pp. 102–112, 2010.
- [19] A. Shintemirov, W. Tang, and Q. H. Wu, "Power transformer fault classification based on dissolved gas analysis by implementing bootstrap and genetic programming," *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, vol. 39, no. 1, pp. 69–79, 2009.
- [20] L. Zhang, J. Lin, Z. Yang, H. Shao, B. Liu, and C. Li, "Wave-ConvNeXt: An Efficient and Precise Fault Diagnosis Method for IIoT Leveraging Tailored ConvNeXt and Wavelet Transform," *IEEE Internet of Things Journal*, vol. 11, no. 13, pp. 23 096–23 109, Jul. 2024.
- [21] M. Alonso-González, V. G. Díaz, B. L. Pérez, B. C. P. G-Bustelo, and J. P. Anzola, "Bearing Fault Diagnosis With Envelope Analysis and Machine Learning Approaches Using CWRU Dataset," *IEEE Access*, vol. 11, pp. 57 796–57 805, 2023.
- [22] X. Zhang, B. Zhao, and Y. Lin, "Machine Learning Based Bearing Fault Diagnosis Using the Case Western Reserve University Data: A Review," *IEEE Access*, vol. 9, pp. 155 598–155 608, 2021.
- [23] Y. Zhao, N. Zhang, Z. Zhang, and X. Xu, "Bearing fault diagnosis based on mel frequency cepstrum coefficient and deformable space-frequency attention network," *IEEE Access*, vol. 11, pp. 34 407–34 420, 2023.

TABLE X: CWRU Bearing Fault Diagnosis Datasets: Configuration, Class Distribution, and Split Homogeneity

Configuration				Sample Distribution				Homogeneity
Dataset	Sampling Rate	Window/Hop	Fault Class	Files	Train	Val	Test	JS Div.
CWRU 12k DE	12 kHz	2048/1024	BF007	4	234	117	118	TV: 0.0021 TT: 0.0020 VT: 0.0000
			BF014	4	236	117	118	
			BF021	4	235	118	118	
			IF007	4	236	117	119	
			IF014	4	234	117	117	
			IF021	4	235	117	118	
			OF007	4	236	118	117	
			OF014	4	234	118	118	
			OF021	4	236	118	118	
			Normal	4	708	473	471	
Total (10 cls)			40	2824	1530	1532		
CWRU 12k FE	12 kHz	2048/1024	BF007	4	233	117	117	TV: 0.0037 TT: 0.0037 VT: 0.0000
			BF014	4	235	118	117	
			BF021	4	233	117	117	
			IF007	4	234	117	117	
			IF014	4	234	117	117	
			IF021	4	234	117	117	
			OF007	4	235	117	117	
			OF014	1	69	22	22	
			OF021	1	69	22	22	
			Normal	4	708	473	471	
Total (10 cls)			34	2484	1337	1334		
CWRU 48k DE	48 kHz	8192/4096	BF007	4	234	118	58	TV: 0.0182 TT: 0.0356 VT: 0.0640
			BF014	4	234	59	117	
			BF021	4	234	117	58	
			IF007	4	234	58	117	
			IF014	4	236	117	14	
			IF021	4	235	118	58	
			OF007	4	235	58	117	
			OF014	4	175	117	118	
			OF021	4	236	59	118	
			Normal	4	175	117	117	
Total (10 cls)			40	2228	938	892		

Note: Ball Fault (BF), Inner Race Fault (IF), Outer Race Fault (OF) with fault diameters 007, 014, 021 mil. Train/Val/Test split: 60/20/20 at file level (seed 42). JS Divergence: TV = Train vs. Val; TT = Train vs. Test; VT = Val vs. Test. JS values ≤ 0.064 indicate excellent distribution homogeneity across all datasets. For 12k FE, OF014 and OF021 have only 1 source file each; windows are drawn from non-overlapping time regions (60%/20%/20%) to populate all splits. Normal recordings are $\sim 4\times$ longer than fault recordings.

- [24] Y. Ruan, X. Yu, Y. He, Y. Fang, J. Wang, and H. Cong, "Hierarchical fault diagnosis method for piezoresistive pressure sensor based on gaf and cnn-svm," *IEEE Transactions on Instrumentation and Measurement*, vol. 74, pp. 1–12, 2025.
- [25] K. Zhao, H. Li, J. Ma, T. Zhuang, Y. Li, H. Xiao, and Z. Yin, "Mechanical fault diagnosis method for gis based on convolution neural network and enhanced gramian angular field," in *2023 5th Asia Energy and Electrical Engineering Symposium (AEEES)*, 2023, pp. 640–643.
- [26] X. Zhai and F. Wang, "Space target recognition based on gramian angular field representation and attention mechanism," in *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium*, 2023, pp. 5774–5777.
- [27] L. Hu, J. Tian, Z. Dong, L. Cui, and C. Lang, "Beyond cnn or transformer alone: A gadf-powered dual-branch network with kan-swin transformer for fault diagnosis of aerospace bearing," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 1047–1063, 2026.
- [28] M. Li, X. Yuan, Z. Luo, and X. Qiu, "Infrared and visual image fusion based on nsst and improved pcnn," *Journal of Physics: Conference Series*, vol. 1069, no. 1, p. 012151, aug 2018.
- [29] Y. Kang, J. Lian, C. Guo, L. Zheng, and M. Zhang, "Enhancement algorithm for color images based on improved FC-MSPCNN," *J. Phys.: Conf. Ser.*, vol. 1961, no. 1, p. 012006, Jul. 2021.
- [30] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 7132–7141.
- [31] Z. Zeng, R. Zhang, W. Cai, and Y. Li, "Application of local histogram clipping equalization image enhancement in bearing fault diagnosis," *IEEE Access*, vol. 10, pp. 49 251–49 264, 2022.